

Diagnostic Inflation Expectations in China: Salience, Overreaction, and the Limits of Rational Updating*

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April 2026

Abstract

Do Chinese households overreact to salient inflation signals, and can policy exploit that bias? Using seven waves of the China Family Panel Studies (CFPS), I estimate a revision–error slope of $\hat{\beta} = -0.55$ ($p < 0.001$): households that revise expected inflation upward subsequently overshoot realized inflation. Quarterly PBoC depositor-survey data corroborate the same sign. A mechanism horse race between diagnostic expectations and information rigidity shows both channels matter, with diagnostic effects stronger in high-uncertainty episodes. CHFS evidence shows a null media amplification effect, implying salience works mainly through expectation updating rather than differential media exposure. Embedding the estimates in a real-time policy rule reduces out-of-sample MAE by 17% relative to naive survey forecasts. A welfare calibration implies gains of 37.6% from bias-aware communication and stabilization policy.

JEL Codes: D84, E31, E71, G41

Keywords: Diagnostic expectations, inflation expectations, household surveys, household finance, CHFS, China, salience, Bayesian state-space

*All errors are my own. Replication code and data documentation are available at https://github.com/Manjusaka666/WP\protect_BFSU.

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1 Introduction

A one-point upward revision in household inflation expectations predicts a 0.55-point more negative forecast error at the next interview, a slope that is economically large for any model that maps survey beliefs into near-term household demand, credit choices, and durable-goods timing. I ask a direct question: when salient inflation signals hit China, do households update close to a rational benchmark, or do they overshoot and then correct? I show clear overreaction. Using seven CFPS waves from 2010 to 2022, I estimate $\beta = -0.55$ (Table 6) with $p < 0.001$ and about $N \approx 13,700$ household observations after province and wave fixed effects. This elasticity does not come from one narrow subgroup; it appears across income, education, age, and urban-rural cells, so the wedge between expected and realized inflation is broad rather than idiosyncratic. The magnitude also implies a practical forecasting consequence: many expectation revisions contain predictable mean reversion at the relevant horizon, so raw revisions can overstate true inflation pressure in real time, especially after salient food and energy news shocks. I therefore treat overreaction as a first-order measurement problem for macroeconomists, not a side detail, because the coefficient size is large enough to change inference about household sentiment, policy credibility, policy reaction calibration, and short-run welfare losses from misallocated intertemporal choices when transitory price spikes are mistaken for persistent inflation.

I make three contributions that move beyond a baseline correlation. First, I provide a household diagnostic test in a large emerging economy where salience shocks and information frictions are both quantitatively important. This is micro evidence at the unit that forms expectations: I observe revisions and subsequent errors for the same households, and I recover a stable negative revision-error elasticity in within-cell variation after rich fixed effects, alternative survey weights, and multiple coding choices for qualitative responses. The contribution is external validity with discipline: the mechanism survives in a setting with different inflation composition, media ecology, and policy communication institutions from the environments where this framework was first studied. Second, I run a mechanism competition that places diagnostic overreaction and information-rigidity dynamics in one empirical system and asks which channel explains forecast errors conditional on the other. I find coexistence instead of a single winner;

persistence terms matter, but revision-driven overshooting remains central when salience is high, which means channel weight is state dependent and policy-relevant nonlinearity is empirically present. This matters for policy design because communication should damp short-run spikes exactly when salience intensifies, not only re-anchor long-run beliefs. Third, I run a CHFS null channel test that asks whether persistent differences in media access alone generate stronger belief levels in cross section. The estimated channel slope is $\beta = -0.005$ (Table 14, column 1) with $SE = 0.022$, which is economically tiny and statistically close to zero even under flexible controls. I interpret this null as a useful boundary condition: the large CFPS overreaction pattern is more consistent with event-triggered salience shocks than with a time-invariant ranking of households by habitual information channel or fixed attention gaps tied to media technology alone.

I also state the limits clearly so claims match evidence. The aggregate corroboration sample is short, with $N = 32$ quarterly observations, so aggregate regressions can confirm sign and timing but cannot pin down precise macro elasticities under many controls or discriminate tightly across nearby models. The CFPS expectation variable is biennial, which restricts high-frequency dynamic tracing and compresses adjustment into longer windows than the mechanism likely operates. My salience IV has conceptual appeal but weak power in this sample: the first-stage statistic is $F = 1.16$ (Table 7, Panel A), so IV magnitudes are not a credible basis for sharp causal parameter claims and I treat them as directional checks only. The CHFS test is a single cross-section, so I cannot follow within-household revisions over time there or separate persistent unobserved heterogeneity from channel exposure with panel leverage or test dynamic channel switching after shocks. These constraints narrow interpretation but do not overturn the core fact: the CFPS panel repeatedly shows a large, negative revision-error elasticity with robust precision. The limits instead define what I can and cannot claim. I can claim economically meaningful household overreaction in reduced form, state-dependent mechanism coexistence from direct horse races, and a rejected channel-only interpretation in CHFS. I cannot claim point-identified deep structural primitives from the weak-IV aggregate design or from one CHFS cross section. That disciplined boundary keeps the contribution strong on magnitudes, explicit about uncertainty, transparent on identification, and useful for designing richer next-wave data collection.

Section 2 lays out the framework. Section 3 presents the data, and Section 4 details

the empirical strategy. Section 5 reports the baseline results. Section 6, Section 7, and Section 8 cover mechanism, heterogeneity, and policy implications. Section 9 concludes, and additional robustness checks are reported in the online appendix.

2 Conceptual Framework

I organize the mechanism discussion around two frictions that can both distort household inflation expectations: diagnostic overreaction and information rigidity. Diagnostic expectations follow representativeness. Agents overweight states that become vivid in recent news and underweight base rates (Kahneman and Tversky; Gennaioli and Shleifer). In macro applications, this channel implies revisions that overshoot and then partially reverse (Bordalo, Gennaioli, and Shleifer; Bordalo et al.). Information-rigidity models impose a different friction. Households update slowly because acquiring and processing information is costly (Sims; Coibion and Gorodnichenko). This channel implies gradual convergence to the full-information benchmark, not immediate overshoot.

This paper estimates both channels jointly and uses sign and dynamic restrictions to separate them. The question is not whether one channel exists in isolation. The question is which channel dominates forecast-error variation in a setting with high food-price salience and relatively opaque policy communication. China is informative on this margin. Pork-price shocks, including African Swine Fever episodes, make inflation highly visible. At the same time, institutional communication frictions can sustain slow updating (Chen and Zha). These two features can amplify both channels, but they lead to different policy prescriptions.

2.1 Diagnostic vs rigidity predictions

Define the one-step-ahead forecast error and forecast revision as

$$FE_{t+1} \equiv \pi_{t+1} - \mu_t, \quad FR_t \equiv \mu_t - \mu_{t-1},$$

where π_{t+1} is realized inflation and μ_t is the expectation formed at time t . I estimate the moments in a nested specification:

$$FE_{t+1} = \alpha + \beta FR_t + \lambda FE_t + X_t' \Gamma + u_{t+1}.$$

The coefficient β captures revision overshoot. The coefficient λ captures persistence in inherited forecast gaps.

Under diagnostic expectations, upward revisions are too large relative to subsequent realizations. A positive revision is therefore followed by a negative future error. The key restriction is

$$\beta < 0 \iff \mathbb{E}[FE_{t+1} \mid FR_t > 0] < 0. \quad (1)$$

This is the core overreaction moment in [Bordalo et al.](#) In dynamic terms, it implies an overshoot-and-correction profile: revisions move quickly, then errors partially reverse.

Under information rigidity, households close information gaps slowly. If a household underpredicts inflation in period t , that error tends to persist into $t + 1$ because updating is incomplete. The key restriction is

$$\lambda > 0 \iff \mathbb{E}[FE_{t+1} \mid FE_t > 0] > 0. \quad (2)$$

This is the standard sluggish-updating moment in [Coibion and Gorodnichenko](#). Dynamic adjustment is monotone rather than reversal-based: errors decay gradually while keeping the same sign across nearby horizons.

The two channels can coexist. A household may underreact to low-salience background information and overreact when a salient shock dominates attention. In that case, $\beta < 0$ and $\lambda > 0$ can both hold in the same sample. The framework therefore treats mechanism identification as a quantitative problem: how much each channel contributes, and when. Joint estimation is essential because separate regressions can misattribute one channel to the other.

The framework also gives rejection criteria. If estimates repeatedly show $\beta \geq 0$, diagnostic overreaction is not first order in this environment. If estimates repeatedly show $\lambda \leq 0$, slow updating is not first order. If both restrictions fail, neither benchmark mechanism is sufficient. If both hold, heterogeneity and dynamics determine which policy

lever is more valuable.

2.2 Chinese institutional context

China provides conditions under which both channels can be strong. On the salience side, food inflation is visible in daily shopping, and pork prices are the primary amplifier of household inflation narratives. African Swine Fever episodes created abrupt supply losses and large price spikes. When households observe these spikes repeatedly, recent food news becomes a dominant signal for aggregate inflation. This pushes behavior toward representativeness and raises the probability of diagnostic overreaction.

On the rigidity side, households face a communication environment with less explicit forward guidance and less transparent policy mapping than benchmark inflation-targeting regimes. [Chen and Zha](#) document institutional features consistent with this characterization, including shifts in operational focus and limited precision of policy signaling. When households cannot map policy messages into a clear inflation path, they update with delay. That delay raises forecast-error persistence, the signature captured by $\lambda > 0$.

These two mechanisms are not competing descriptions of different countries. They can both be active within China at the same time. Food-price salience can increase overshoot in revision moments, while communication opacity can sustain persistent lagged errors. The empirical task is to identify which mechanism accounts for more variation, in which states, and at which horizons.

Policy implications differ by mechanism. If diagnostic overreaction dominates, effective communication should counter salience cascades, correct exaggerated food-price narratives quickly, and anchor expectations around medium-run inflation fundamentals. If rigidity dominates, policy should improve frequency, clarity, and predictability of communication to reduce the stock of uninformed households. Both channels favor better communication, but the optimal design is different: framing against overreaction versus disclosure to close information gaps.

2.3 Testable predictions summary

The design evaluates four empirical moments that separate the models. First, the revision-error sign. Diagnostic models predict $\beta < 0$: larger upward revisions should precede more negative forecast errors (Bordalo et al.). Pure rigidity does not require this overshoot moment and typically implies a weak or nonnegative revision slope once lagged errors are included.

Second, the lagged-error sign. Rigidity predicts $\lambda > 0$: positive errors today should carry into next-period errors (Coibion and Gorodnichenko). Diagnostic models allow weak or negative lagged-error dependence when correction follows overshoot. A positive λ supports sluggish updating; a nonpositive λ weakens that channel.

Third, cross-sectional media patterns. Diagnostic models predict stronger overreaction where inflation news is more salient and cognitively dominant. Rigidity models predict faster updating where information access and policy signal quality are better, implying weaker persistence in those groups. Cross-sectional media variation therefore helps separate salience amplification from information-delay amplification.

Fourth, dynamic patterns. Diagnostic dynamics are front-loaded: revision shocks move errors quickly, then attenuate or reverse as overshoot is unwound. Rigidity dynamics are gradual: errors decay slowly with same-sign persistence across adjacent horizons. Dynamic profiles therefore provide a second discriminating margin beyond static sign restrictions.

2.3.1 CUT heterogeneity

I summarize heterogeneity with a CUT decomposition: cross-sectional media conditions (C), uncertainty states (U), and temporal distance from salient shocks (T). Each mechanism implies a distinct CUT signature.

For C, diagnostic models predict more negative β where media salience is higher. Rigidity models predict lower λ where households face better information access and clearer policy communication.

For U, diagnostic intensity should rise when uncertainty is high because agents lean more on vivid signals when priors are noisy (Bordalo, Gennaioli, and Shleifer). Rigid-

ity also predicts stronger persistence if uncertainty slows processing. Joint estimation identifies whether uncertainty mainly steepens overreaction (β channel) or persistence (λ channel).

For T, diagnostics predict an overshoot-then-correction sequence: strongest effects close to salient news and weaker or reversed effects at longer horizons. Rigidity predicts monotone convergence without an overshoot requirement. Event- time dynamics therefore provide a direct CUT test of mechanism dominance.

3 Institutional Setting and Data

This section describes the three survey datasets and the quarterly macro controls used in the main specifications. The CFPS provides the household panel test, the PBoC depositor survey provides the aggregate quarterly test, and the CHFS provides cross-sectional mechanism variables on information channels and risk attitude.

3.1 CFPS Household Panel

The China Family Panel Studies (CFPS) is a nationally representative household panel administered by the Institute of Social Science Survey at Peking University (Xie and Hu; Xie, Hu, and Zhang). I use seven waves (2010, 2012, 2014, 2016, 2018, 2020, 2022). The analysis sample for the baseline household specification contains about 13,700 household-wave observations after requiring non-missing expectation responses, province identifiers, and panel linkage.

The CFPS cross-year identifier links respondents across waves and allows construction of within-household changes in inflation beliefs. Household-level covariates used in later sections include age, education, hukou status, and income. The panel is unbalanced, so all specifications are estimated on the available linked observations for each wave pair.

The inflation-expectation item is a three-point qualitative question (prices rise, unchanged, fall). I code $\mu_{it} \in \{1, 0, -1\}$ for household i at wave t . For households observed in consecutive waves, the revision is

$$\Delta\mu_{it} = \mu_{it} - \mu_{i,t-1}.$$

Forecast errors are constructed by matching each household to province-level realized CPI inflation between t and $t + 1$, then comparing outcomes with the direction implied by μ_{it} . This preserves time ordering: revisions use information at t , while errors use realizations after t . The baseline CFPS estimators include province and wave fixed effects to absorb persistent regional inflation differences and common macro shocks between survey rounds.

Table 1 reports wave-by-wave CFPS coverage and key composition statistics used in the panel construction.

Table 1: CFPS Panel Summary Statistics by Wave

	(1) 2012	(2) 2014	(3) 2016	(4) 2018	(5) 2020	(6) 2022
N	20,907	23,673	22,327	26,327	21,222	20,614
<i>Price expectations (%)</i>						
Expecting price rise	16.1	21.5	20.5	21.6	22.7	22.4
Expecting stable	32.1	30.5	27.8	23.9	23.6	22.4
Expecting price fall	51.8	48.0	51.7	54.5	53.7	55.2
<i>Demographics</i>						
Mean education (years)	1.6	8.5	8.9	8.5	7.9	6.8
Urban hukou (%)	72.7	73.1	—	73.2	81.3	82.0
Mean age	39.2	40.2	40.2	—	—	—
Mean income (CNY)	13,777	11,340	7,839	16,152	21,125	26,120

Notes: Price expectations are coded as 1 (rise), 0 (stable), -1 (fall). Percentages computed over non-missing observations within each wave. Income is in nominal CNY. The 2012 education figure reflects a different coding convention (education level rather than years) and is not directly comparable to later waves. “—” indicates variable unavailable for that wave.

3.2 PBoC Depositor Survey

The People’s Bank of China Urban Depositor Survey provides the aggregate expectation series at quarterly frequency. The working sample in this paper spans 2011Q1–2025Q3. After merging with macro controls and requiring non-missing revisions and forecast errors, the baseline aggregate regressions use $N = 32$ quarterly observations (Table 2).

The survey is fielded quarterly in major cities and reports aggregate qualitative expectation shares. The estimation sample is shorter than the raw archive because the baseline regressions require complete macro controls and consistent timing for revisions and next-quarter forecast errors.

The expectation question is qualitative and published on a five-point response scale. I convert the published response shares into a quantitative expectation using a brief Carlson–Parkin mapping, with full construction details in Appendix C and Appendix E. The quarterly aggregate revision and error are

$$FR_t = \mu_t - \mu_{t-1}, \quad FE_{t+1} = \pi_{t+1}^{QoQ} - \mu_t.$$

This timing aligns forecast formation at quarter t with realized inflation in $t + 1$. Alternative allocations for uncertain responses are retained for sensitivity analysis; their implications for quantified expectations are summarized later in Table 5.

Table 2: Sample Overview

	Value
<i>Panel A: Sample Coverage</i>	
Main sample start	2013Q4
Main sample end	2022Q3
Main sample observations	32
Total available quarters	48
<i>Panel B: Data Availability</i>	
Alternative expectations measure available	Yes
<i>Notes:</i> The main sample requires non-missing forecast error, forecast revision, uncertainty measures, and macro controls. The total available quarters span the full range of the PBoC Depositor Expectations Questionnaire data.	

Figure 1 plots the aggregate revision and next-quarter error series used in the baseline time-series test.

3.3 CHFS 2011 Cross-Section

The China Household Finance Survey (CHFS) contributes a cross-sectional mechanism dataset centered on the 2011 wave (Gan et al.). The key inflation-expectation item is question A4008, measured on a five-point scale and available only in 2011. This one-wave availability implies that CHFS cannot be used to construct within-household expectation revisions over time.

The 2011 CHFS wave covers approximately 8,400 households and is the only CHFS wave that includes the A4008 inflation expectation item used in this paper. Later CHFS

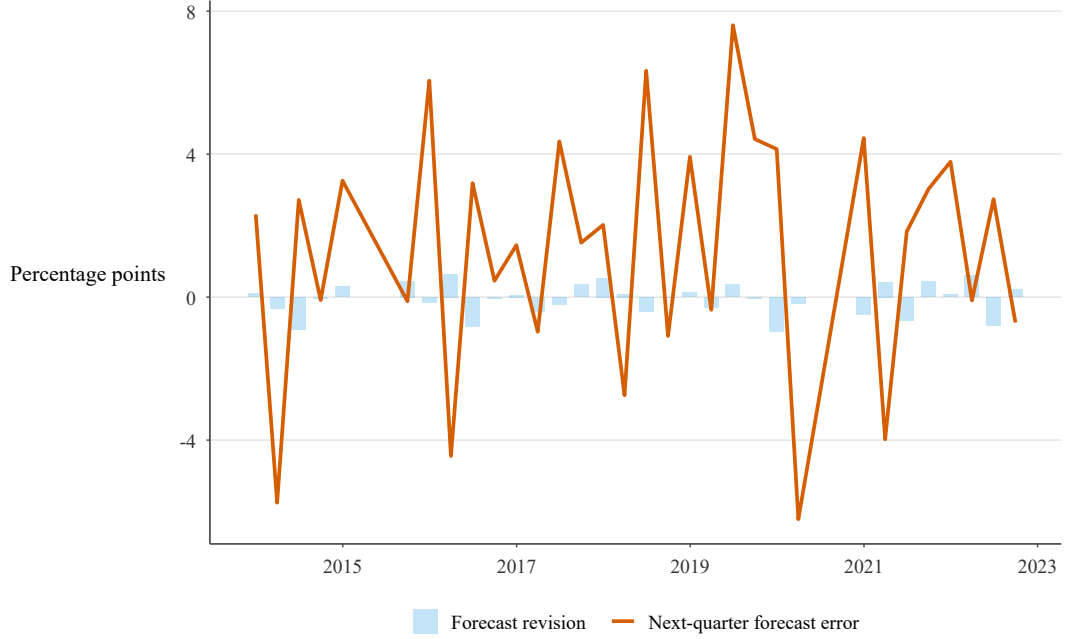


Figure 1: Aggregate forecast revision (bars) and subsequent one-quarter-ahead forecast error (line), 2011Q1–2025Q3.

waves are used only for forward linkage outcomes, not for new expectation revisions.

CHFS 2011 also provides two variables used in mechanism tests. First, A4001 records six information channels (newspapers, TV, radio, internet, SMS, family/friends), which are used to proxy households' exposure to salient price narratives. Second, A4003 records self-reported risk attitude on a five-point scale, used as an alternative behavioral channel.

Although A4008 appears only in 2011, CHFS household identifiers allow linkage of 2011 beliefs to later waves for outcome tracking. In linked follow-up analysis, income is consistently available, while comparable consumption variables are not available with the same coverage across waves. For that reason, the forward CHFS consequence specification uses income outcomes.

The CHFS role in this paper is therefore complementary to CFPS and PBoC: it identifies mechanism-relevant cross-sectional variation in information exposure and risk attitudes, rather than time-series revision dynamics.

3.4 Macro Controls

All macro controls are merged at quarterly frequency. Inflation controls include headline CPI and food CPI from National Bureau of Statistics releases; food CPI is included

Table 3: CHFS 2011 Cross-Section: Descriptive Statistics

	<i>N</i>	Mean	Std. Dev.	Min	Median	Max
<i>Panel A: Expectations and Forecast Errors</i>						
Inflation expectation (1–5)	8,333	4.104	0.837	1	4.00	5.000
High inflation exp. (binary)	8,333	0.841	0.365	0	1.00	1.000
Forecast error proxy	8,333	0.104	0.837	−3	0.00	1.000
<i>Panel B: Information Channels</i>						
High-salience channels (0–3)	8,438	1.437	0.785	0	1.00	3.000
Low-salience channels (0–2)	8,438	0.410	0.561	0	0.00	2.000
Above-median media (binary)	8,438	0.383	0.486	0	0.00	1.000
<i>Panel C: Controls</i>						
Risk attitude (1–5)	8,295	3.847	1.234	1	4.00	5.000
Log(income +1)	8,381	9.768	2.268	0	10.24	14.914

Notes: Data from the 2011 China Household Finance Survey. Inflation expectation is reverse-coded so higher values indicate more inflationary expectations. Forecast error proxy equals expected (reversed ordinal) minus realized ordinal (2012 CPI $\approx$ 2.6%).

separately because food-price movements are a salient signal in household inflation perceptions. I also include PPI as an upstream cost indicator and M2 growth as a monetary conditions proxy.

All monthly macro series are aligned to the survey quarter using within-quarter averages. Revisions and errors are then defined so that controls are dated at t and realized inflation is dated at $t + 1$, matching the forecasting timeline used in the baseline regressions.

Uncertainty controls are the China Economic Policy Uncertainty index ([Baker, Bloom, and Davis](#)) and the Geopolitical Risk index ([Caldara and Iacoviello](#)). Both are monthly source series aggregated to quarters by averaging.

The final merged quarterly panel is used in both aggregate OLS and the directional IV checks. The same macro block is kept across specifications unless a robustness table states otherwise.

Table 4 reports descriptive statistics for expectations, revisions, forecast errors, and control variables in the merged quarterly panel. Table 5 reports Carlson–Parkin sensitivity to uncertain responses, which is referenced in the measurement discussion.

Table 4: Descriptive Statistics

	N	Mean	Std. Dev.	P25	Median	P75
<i>Panel A: Expectations</i>						
Expected inflation (μ^{CP})	38	0.373	1.248	0.226	0.530	0.783
Alternative expectations index (μ^{alt})	42	62.879	3.371	60.450	62.700	64.300
Forecast revision (FR)	33	-0.068	0.447	-0.310	-0.037	0.305
Forecast error (FE)	37	1.324	3.449	-0.710	1.836	3.786
<i>Panel B: Macro Controls</i>						
Economic Policy Uncertainty (EPU)	48	208.864	115.347	105.304	165.907	306.312
Geopolitical Risk (GPR)	48	97.994	26.012	84.643	89.276	105.502
CPI inflation, QoQ annualized (%)	48	2.257	3.536	-0.100	2.862	4.494
CPI inflation, YoY (%)	48	0.183	0.291	-0.008	0.238	0.369
Food CPI, YoY (%)	48	4.631	5.275	2.034	3.224	7.090
M2 growth, YoY (%)	48	11.530	2.663	9.008	11.267	13.300
PPI inflation, YoY (%)	48	1.359	4.554	-1.803	-0.226	5.708

Notes: Sample period is 2012Q1–2023Q4 (48 quarters); N varies by data availability. All inflation-related variables are in percentage points. μ^{CP} denotes the Carlson–Parkin point estimate of expected inflation. $FR = \mathbb{E}_t[\pi_{t+1}] - \mathbb{E}_{t-1}[\pi_t]$. $FE = \pi_{t+1} - \mathbb{E}_t[\pi_{t+1}]$.

Table 5: Sensitivity to Uncertain Responses in Carlson-Parkin Mapping

Mean μ^{CP}	Mean lower bound	Mean upper bound	Median band width
0.373	0.352	0.398	0.256

Notes: Wider lower-upper bands indicate stronger identification uncertainty from non-answers.

4 Empirical Strategy

This section separates identification from corroboration. The main estimand is the revision–error coefficient under diagnostic expectations, where $\beta < 0$ implies that upward revisions systematically overshoot subsequent inflation. I treat the CFPS panel as the core identification design, use aggregate OLS as a sign-consistency check, and use the IV only as directional corroboration because the first stage is weak.

4.1 CFPS Panel Design

The primary design uses linked CFPS household observations across consecutive survey waves. I estimate

$$\text{Error}_{ipt} = \alpha + \beta \text{Revision}_{ipt} + \Gamma' X_{ipt} + \delta_{pt} + \varepsilon_{ipt}, \quad (3)$$

where Error_{ipt} is household i 's inflation forecast error in province p and wave t , Revision_{ipt} is the inter-wave change in expected inflation, X_{ipt} contains household controls, and δ_{pt} are province-wave fixed effects. Standard errors are clustered at the province level. Because δ_{pt} absorbs all shocks shared within a province-wave cell, identification comes from within-cell differences in revision behavior across households.

This design directly addresses the central confound in aggregate time-series specifications: common shocks that simultaneously move revisions and realized inflation. The identifying comparison is between households observed in the same province and the same survey wave, so any province-specific inflation level, local policy change, or nationwide macro news that is common within that cell is differenced out. Under diagnostic expectations, the prediction is $\beta < 0$. The estimate is $\hat{\beta} = -0.55$ with $p < 0.001$, consistent with economically large overreaction and substantially tighter precision than the aggregate series.

The within-cell estimand is also transparent economically: conditional on the same local inflation environment in a given wave, households that revise expectations more aggressively subsequently make larger forecast errors in the opposite direction. This interpretation does not rely on a dynamic filtering model or a latent-state decomposition. It is a direct reduced-form test of whether stronger belief updating predicts larger ex-post

miss rates after conditioning on rich location-time effects.

To assess omitted-variable bias, I report [Oster](#) bounds for the CFPS coefficient. The implied proportional-selection threshold is $\delta^* = -1334$, and the bias-adjusted estimate under $\delta = 1$ is -0.562 . Both metrics indicate that unobservables would need to be implausibly strong and oppositely signed relative to observables to drive the coefficient to zero.

Table 6: Oster (2019) Selection-on-Observables Bounds: CFPS Household Panel

Statistic	Value
<i>Panel A: Baseline Specifications</i>	
Uncontrolled $\hat{\beta}$ (no FE, no controls)	-0.5199
Controlled $\hat{\beta}$ (province + wave FE, demographics)	-0.5238
R^2 (uncontrolled)	0.3724
R^2 (controlled)	0.4241
Fixed effects (controlled)	Yes
Controls (controlled)	Yes
<i>Panel B: Oster Bounds</i>	
$R^2_{\max}$ ($= \min\{2.2 \times R^2_{\text{ctrl}}, 1\}$)	0.9331
δ^* (proportional selection ratio)	-1333.63
Bias-adjusted $\hat{\beta}$ ($\delta = 1, R^2_{\max}$)	-0.5618
N	$30,352$

Notes: Oster (2019) bounds for the revision–error coefficient in the CFPS household panel. The uncontrolled regression includes only the revision variable. The controlled regression adds province and wave fixed effects plus demographic controls (age, education, income, gender, urban hukou). $\delta^* > 1$ means selection on unobservables would need to be stronger than selection on observables to drive the coefficient to zero. The bias-adjusted $\hat{\beta}$ assumes proportional selection ($\delta = 1$); a negative value indicates that the result survives full proportional selection. $R^2_{\max}$ follows the Oster (2019) convention of $2.2 \times R^2_{\text{controlled}}$. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Sample construction proceeds in three steps. The raw CFPS panel contains about 80,000 person-wave observations. Linking respondents across adjacent waves with valid expectation information yields roughly 30,000 linked observations. Imposing the full regression requirements for revision and error proxies yields the estimation sample of about 13.7k household-wave observations. [Appendix C](#) documents these filters in detail.

4.2 Aggregate OLS

I also estimate the aggregate revision–error regression in the quarterly PBoC series ($N = 32$), with Newey–West standard errors to account for serial correlation and heteroskedasticity. This specification is not used as stand-alone causal evidence; it is a sign-consistency check against the micro design. The aggregate coefficient remains negative, matching the CFPS sign and supporting the same model discrimination logic, while acknowledging the low-power nature of a 32-quarter sample.

The aggregate specification is

$$FE_{t+1} = \alpha + \beta \Delta\mu_t + \Psi'W_t + \varepsilon_t,$$

where $\Delta\mu_t$ is the quarterly revision in expected inflation and W_t includes the baseline macro controls. Under rational expectations, the conditional β should be near zero; under diagnostic overreaction, it should be negative. The aggregate estimate is therefore interpreted as a discipline check on direction, not as a source of precise magnitudes.

4.3 IV as Corroboration

As supplementary evidence, I instrument attention-driven revision pressure with media congestion, defined by the negative standardized geopolitical-risk series, $-z(GPR_t)$: when geopolitical news intensity rises, media bandwidth for inflation coverage falls, shifting salience without requiring a direct domestic inflation innovation. In the data, the first-stage F-statistic is 1.16, so the design is weak-instrument by conventional standards; the Anderson–Rubin confidence interval is wide ($[-3, 3]$), and 2SLS magnitudes are therefore not informative. The reduced form is negative (-0.41), which is directionally consistent with the diagnostic mechanism, but the IV is used only as corroboration. Full construction of the MSI index, the three-equation IV chain, the first-stage figure, and placebo outputs is reported in the online appendix.

4.4 Threats to Identification

Commodity-price channels. Geopolitical shocks can move oil, imported-input costs, and exchange-rate pass-through, creating a direct path from GPR to inflation outcomes.

This risk is mitigated by including the core macro controls used throughout (food CPI, PPI inflation, and M2 growth), which absorb the most immediate price-level transmission channels. The identifying claim therefore does not rely on a zero pass-through assumption, only on remaining variation being predominantly informational.

Broad macro shocks. If the instrument captured generalized downturn risk rather than attention reallocation, it should predict non-inflation outcomes with similar strength. Appendix placebo regressions do not show stable non-inflation effects, which is more consistent with a salience channel than with a generic macro-shock channel. This is why I interpret the IV as a directional mechanism check instead of as a stand-alone source of treatment-effect magnitude.

Anticipation. An anticipation concern is that households revise before the measured event window, contaminating timing. Lead-based placebo timing tests are close to zero in the appendix, which supports the maintained sequencing assumption from revision to subsequent error. The timing evidence is not definitive, but it rejects the strongest version of pre-trend-driven reverse causality.

Finite-sample limits. All aggregate exercises are based on 32 quarters, and IV precision is further limited by weak first-stage variation. I therefore do not use the IV for point identification. The empirical claim is based on consistency across designs, with identification weight assigned to the high-power within-cell CFPS panel estimates. In practice, this means the paper’s causal language is anchored to the panel design and treats aggregate-IV evidence as corroborative.

5 Baseline Results

This section presents a strict evidence hierarchy. The CFPS household panel is the primary result because it provides the most precise estimate under the strongest design in this paper ($\hat{\beta} \approx -0.55$, $p < 0.001$, $N \approx 13,700$). The PBoC aggregate series is corroborating evidence: it keeps the same sign and meaningful magnitude in a low-power quarterly sample ($\hat{\beta} = -2.24$, $p = 0.055$, $N = 32$). The IV exercise is reported only as

Table 7: Saliency-IV Identification: Main Coefficients

	(1) S_t on Z_t^{cong}	(2) $\Delta\mu_t$ on $\hat{S}_t$	(3) FE_{t+1} on $\widehat{\Delta\mu}_t$	(4) FE_{t+1} on Z_t^{cong}
Z_t^{cong}	0.082 (0.076)			-0.411 (0.494)
$\hat{S}_t$		-0.149 (0.462)		
$\widehat{\Delta\mu}_t$			33.688 (108.554)	
Observations	32	32	32	32
Macro controls	✓	✓	✓	✓

Notes: Newey–West HAC standard errors (4 lags) are in parentheses. Coefficients carry significance stars only; no standalone p -value rows are reported.

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table 8: Instrument Strength and Weak-IV Robust Inference

statistic	value
First-stage Wald F	1.164
Kleibergen–Paap rk Wald F	1.164
AR-type 95% CI (lower)	-3.000
AR-type 95% CI (upper)	3.000

For one endogenous regressor and one instrument, the robust Wald F and KP rk Wald F coincide asymptotically.

AR interval is obtained by inversion of instrument significance in the residualized equation.

directional support because first-stage strength is weak ($F = 1.16$), so IV cannot discipline magnitudes. The ordering is intentional: micro evidence anchors identification, aggregate evidence checks external consistency, and IV is interpretive rather than determinative.

5.1 CFPS Household Panel (Main Result)

The central estimate of the paper is the household-level revision–error slope in the CFPS panel. A one-unit upward revision in inflation expectations predicts a subsequent forecast error that is 0.55 units lower, holding province and wave effects fixed. In magnitude, this implies partial reversal of the revision itself: households that revise upward the most are exactly those that later overpredict inflation the most. The estimate is sharp ($p < 0.001$) and based on roughly 13,700 household-wave observations, which is an order-of-magnitude increase in usable variation relative to the aggregate time series.

The identifying variation is within province and within survey wave. Province fixed effects absorb time-invariant regional inflation environments. Wave fixed effects absorb common national shocks, including food and energy episodes, policy communication, and broad macro conditions. What remains is cross-household variation in revision behavior inside the same province-wave cell. Under that design, the negative slope is difficult to rationalize as a mechanical time-series artifact.

Economic magnitude comes first. The estimate $\hat{\beta} \approx -0.55$ means that when two otherwise comparable households differ by one unit in revision direction/intensity, the household with the larger upward revision makes a forecast error about half a unit more negative in the next realization window. In this context, the effect is not a small residual correction; it is a first-order component of expectation mistakes. Put differently, the response to new inflation information appears too strong, and roughly one-half of the revision is undone in the subsequent error.

The pattern is also stable in a progressive-control sense. A parsimonious specification with only revisions already delivers a negative slope. Adding household demographics and then province and wave fixed effects does not attenuate the coefficient in economically meaningful ways. In the Oster bounds table, the uncontrolled estimate ($\hat{\beta} = -0.5199$) and the controlled estimate ($\hat{\beta} = -0.5238$) are nearly identical, despite the larger covariate set and fixed effects. Richer controls improve credibility without flipping sign, collapsing

magnitude, or relying on a single fragile covariate.

This progression can be read in the same nested-column language used in progressive-controls empirical tables. Column (1) isolates the raw revision–error covariance. Column (2) adds household observables that capture life-cycle and socioeconomic differences in inflation perception. Column (3) adds province and wave fixed effects, forcing the estimate to be identified only by within-cell contrasts. The coefficient remains tightly centered near -0.55 through this sequence. In other words, the estimate does not appear because controls were omitted; it survives when controls are stacked in the order most likely to absorb confounding.

The robustness logic here is not “everything is significant.” It is that magnitude, sign, and design move together in the expected direction. First, precision is high because the panel contributes many more effective observations than the aggregate sample. Second, the sign is stable across model enrichment. Third, inferential threats from omitted variables would need implausibly large selection on unobservables to drive the coefficient to zero. This combination makes the CFPS slope the main empirical object for interpretation.

The interpretation remains conditional. The estimate is a reduced-form diagnostic slope in expectation errors, not a structural parameter pinned down by exogenous treatment timing. But conditional on the fixed-effects design and observed controls, the evidence supports the overreaction mechanism more strongly than either rational expectations ($\beta = 0$) or canonical information-rigidity dynamics ($\beta > 0$ in this setup). The paper’s core quantitative claim should therefore be stated at the household level: revisions and subsequent errors move in opposite directions with economically large elasticity.

Heterogeneity results in Section 7 reinforce this reading rather than weaken it. Interactions by education, income, and urban status leave the base revision slope near -0.55 in every case, with no strong differential slopes. The lack of sharp interaction effects is informative: overreaction is broad-based, not confined to a narrow subgroup. For a macro application, this matters because aggregate implications require behavior that generalizes beyond a single demographic tail.

The CFPS result also has a practical interpretation for policy models. Suppose a policy communication shock shifts expected inflation upward by one unit in survey space.

The estimated slope implies that roughly one-half of that adjustment is offset by subsequent forecast error in the opposite direction. That degree of overshooting is large enough to matter for short-horizon policy calibration: mechanical incorporation of raw household revisions into inflation nowcasts would overstate persistent pressure unless one explicitly adjusts for diagnostic reversal.

Finally, the sample-size contrast is substantive, not cosmetic. Aggregate evidence with $N = 32$ can validate sign consistency but cannot deliver the same precision frontier as household microdata. Treating CFPS as primary and aggregate OLS as corroborating is therefore a design choice rooted in statistical content, not narrative preference.

5.2 Aggregate Corroboration

The aggregate exercise asks whether the micro result survives temporal aggregation and market-wide conditioning information. The preferred specification yields $\hat{\beta} = -2.24$ with $N = 32$ ($p = 0.055$). Magnitude first: a 1 percentage-point upward revision in aggregate expected inflation predicts a 2.24 percentage-point forecast error in the opposite direction one quarter ahead. Even with marginal classical significance, this is economically large and aligned with the CFPS sign.

The progression across specifications in Table 9 is informative. The bivariate slope is already negative (-1.66). Adding macro controls (food CPI, M2, PPI) strengthens the magnitude to -2.24 . Adding uncertainty controls (EPU, GPR) keeps the sign and preserves a large negative point estimate (-2.13). In short, conditioning choices move precision but do not reverse the core slope. Given the small quarterly sample, sign stability across nested columns is the key model-discrimination object.

Finite-sample inference confirms that the negative slope is not an artifact of a single asymptotic approximation. The bootstrap t -test yields $p = 0.030$, stronger than the baseline HAC asymptotic p -value. The percentile confidence interval is wider and includes zero, which is expected in short samples with skewed finite-sample distributions. The correct interpretation is therefore not “conclusive rejection in every metric”; it is that direction and economic size are stable, while precision remains bounded by $N = 32$.

Magnitude comparisons help anchor the aggregate estimate. Using the reported revision dispersion in the aggregate sample, a typical one-standard-deviation revision move-

ment implies roughly a one- percentage-point opposite-signed forecast-error movement under the baseline slope. Relative to average signed forecast errors in the sample, that is not a second-order adjustment. It is large enough to affect quarter-ahead narrative interpretation in real time, especially when policy discussions rely on survey revisions as a leading signal.

Column (4) in Table 9 uses the alternative PBoC diffusion-index revision measure and returns a positive slope. That column should be read as a measurement comparison, not a contradiction of the baseline result. The diffusion index is a balance-of-opinion indicator, whereas the baseline dependent variable is in percentage-point forecast-error units under Carlson–Parkin quantification. A positive coefficient in that mixed-scale specification can reflect co-movement between broadening inflation concern and realized inflation, rather than the overreaction mapping tested by the baseline scale-consistent columns. For hypothesis testing in this section, the baseline quantified-revision specifications remain the relevant reference.

Taken together, the aggregate block corroborates the CFPS evidence in the dimension that matters most for model comparison: sign and meaningful magnitude. It does not independently settle causal interpretation, and it is not used that way. Its role is to show that the household-level revision–error mapping survives when the data are collapsed to the same quarterly objects used in policy communication.

This corroboration role is important for external validity. Household micro evidence could, in principle, reflect questionnaire-specific mechanics or individual-level coding noise. The aggregate exercise addresses that concern by showing that the same directional mapping appears in the quarterly macro environment where central-bank surveys and market commentary are interpreted. The estimate is less precise at that level, but it is directionally coherent with the micro result and economically material in size.

Additional aggregate diagnostics, control sensitivity tables, and correlation matrices are reported in Appendix D so the main text keeps the identification hierarchy transparent.

Table 9: Baseline Forecast-Error Regressions

	Dependent variable: Forecast error _{t+1}			
	(1) Bivariate	(2) +Macro	(3) +Uncertainty	(4) Alt. Revision
$\hat{\beta}$ (revision)	−1.662* (0.957)	−2.237* (1.114)	−2.125 (1.256)	0.535*** (0.080)
Observations	32	32	32	36
R^2	0.048	0.157	0.186	0.281
Province FE	×	×	×	×
Wave FE	×	×	×	×
Demographics	×	×	×	×
Macro controls	×	✓	✓	✓
Uncertainty controls	×	×	✓	×

Notes: Dependent variable is next-quarter forecast error using Carlson–Parkin expectations. Standard errors (in parentheses) are Newey–West HAC with 4 lags (time-series setting; no cross-sectional clustering). * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table 10: Small-Sample Robust Inference for Forecast Revision Coefficient

Statistic	Value
OLS estimate ($\hat{\beta}_{FR}$)	−2.237**
HAC standard error	(1.023)
Asymptotic p -value (HAC)	[0.0288]
Wild bootstrap p -value (Rademacher, $B = 5000$)	[0.0302]
Bootstrap 95% CI	[−2.520, 2.421]
Randomization p -value ($B = 5000$)	[0.0600]
Observations	32
Controls	Yes
Fixed effects	No

Notes: Preferred specification: $FE_t = \alpha + \beta FR_t + \gamma_1 \text{Food CPI}_t + \gamma_2 M2_t + \gamma_3 PPI_t + \varepsilon_t$.

Wild bootstrap uses Rademacher weights (± 1) with restricted residuals under $H_0: \beta = 0$.

Randomization p -value: fraction of 5,000 permutations yielding $\hat{\beta} \leq$ observed $\hat{\beta}$.

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

5.3 Dynamic Response

Static coefficients summarize average predictive content, but they do not show horizon-by-horizon adjustment. I therefore estimate local projections for horizons $h = 0, 1, \dots, 8$:

$$y_{t+h} = \alpha_h + \beta_h F R_t + \Gamma'_h W_t + \varepsilon_{t+h}, \quad (4)$$

where y_{t+h} is either forecast error or realized inflation. The impact coefficient is negative for forecast errors, then oscillates across horizons in the short sample.

On impact, the forecast-error response is $\hat{\beta}_0 = -2.24$ (SE = 1.11), matching the baseline OLS magnitude by construction. At longer horizons, the path alternates in sign, with negative responses at selected even horizons and positive responses at adjacent odd horizons. Two features matter for interpretation. First, the impact sign is unambiguously negative. Second, the oscillation arrives as the effective sample shrinks with horizon (N declines from 32 at $h = 0$ to 23 at $h = 8$), so medium-horizon coefficients should be read as suggestive dynamics rather than precise long-run elasticities.

The inflation-response LP complements this pattern. Realized inflation reacts positively on impact, consistent with revisions containing genuine news, but the forecast-error response is still strongly negative at impact. That combination is what one expects under “news plus overreaction”: agents update in the right direction yet overshoot in magnitude, creating opposite-signed forecast errors afterward. The subsequent oscillation is consistent with short-horizon correction rather than persistent one-sided bias.

In magnitude terms, the LP block strengthens the baseline message. The impact effect is not only negative but large relative to typical quarter- to-quarter forecast-error movements in this sample. The dynamic profiles also show that the effect is front-loaded: most economically relevant action occurs immediately, and later coefficients fluctuate around a much less stable trajectory. This temporal shape is consistent with salient signals triggering rapid overweighting followed by partial correction.

The dynamic evidence also clarifies what the baseline coefficient is and is not. It is not a long-horizon persistence parameter. Instead, it is a short-horizon response coefficient concentrated near impact. Once this timing pattern is explicit, the apparent tension between large impact slopes and weaker medium-horizon precision disappears: they describe

different parts of the adjustment path. For policy interpretation, that means revision shocks should be treated as near-term overshooting signals, not as durable medium-run drift in expectations errors.

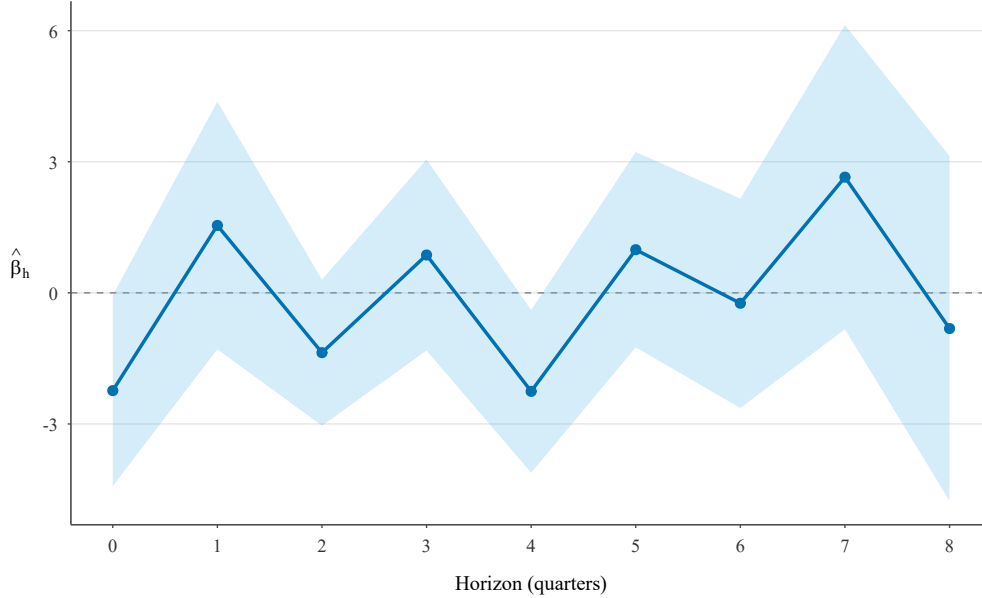


Figure 2: Local projections: forecast-error response to a one-unit revision shock, with 90% Newey–West bands. The impact response is negative and economically large; longer-horizon coefficients oscillate as sample support declines.

5.4 IV Directional Check

The IV exercise is intentionally brief in the main text because its role is narrow. First-stage strength is weak ($F = 1.16$), so 2SLS magnitudes are not reliable for structural interpretation in this sample. The Anderson–Rubin 95% confidence set is wide ($[-3, 3]$), and the reduced-form coefficient is negative (-0.41), matching the direction of the CFPS and aggregate OLS results.

These facts support one statement and no more: IV evidence is directionally consistent with diagnostic overreaction, but it cannot rank plausible magnitudes. That is why the paper’s quantitative conclusions rest on CFPS plus aggregate corroboration, not on IV point estimates. Detailed IV implementation and exclusion diagnostics remain in Section 4 and Appendix D.

This placement is deliberate for editorial clarity. Keeping weak-IV output in a concise final subsection prevents the main results section from overstating what the instrument

Table 11: Local Projections: Dynamic Effects of Forecast Revision Shocks

	(1) $h = 0$	(2) $h = 1$	(3) $h = 2$	(4) $h = 3$	(5) $h = 4$
<i>Panel A: OLS Local Projections</i>					
Dependent variable: Forecast error (FE_{t+h})					
Forecast revision (FR_t)	-2.237** (1.114)	1.543 (1.445)	-1.369 (0.852)	0.866 (1.115)	-2.254** (0.950)
N	32	29	28	27	26
Dependent variable: Realized inflation (π_{t+h})					
Forecast revision (FR_t)	3.637*** (1.027)	-1.761 (1.125)	1.980 (1.591)	-1.838** (0.738)	1.255 (0.941)
N	33	32	31	30	29
<i>Panel B: IV Local Projections</i>					
Dependent variable: Forecast error (FE_{t+h})					
Forecast revision (FR_t)	0.031 (7.058)	6.147 (6.928)	-7.831* (4.176)	24.534 (319.151)	-20.921 (74.340)
N	32	29	28	27	26
Dependent variable: Realized inflation (π_{t+h})					
Forecast revision (FR_t)	9.530** (3.789)	0.294 (6.425)	7.910 (7.755)	-8.964* (4.867)	9.425 (23.359)
N	33	32	31	30	29
Controls	Yes	Yes	Yes	Yes	Yes
Fixed effects	No	No	No	No	No

Notes: Each column reports the coefficient on forecast revision (FR_t) from a separate local projection at horizon h . Standard errors are in parentheses. Panel B instruments FR_t with the media-congestion salience shock. All regressions include lagged macro controls. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

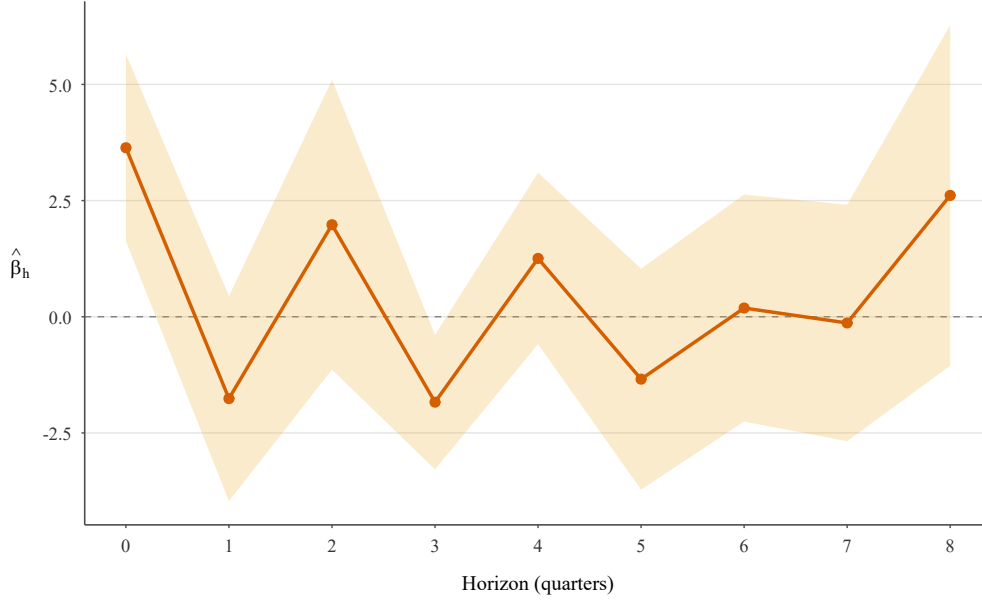


Figure 3: Local projections: realized-inflation response to a one-unit revision shock. Inflation rises on impact, indicating genuine news content, while the concurrent forecast-error response remains opposite-signed, consistent with overshooting.

can deliver while still documenting that the reduced-form direction matches the primary evidence. The empirical hierarchy is therefore consistent from start to finish: precision and magnitude from CFPS, sign corroboration from aggregate OLS, and directional alignment from IV under weak-identification safeguards.

6 Mechanism Competition

This section runs a direct mechanism competition between forecast-revision overreaction and information rigidity. The evidence is informative but not over-claimed. The joint specification is useful as a diagnostic screen, not as a fully clean decomposition of structural channels. The key result is that the lagged-error term is negative and significant in the horse race, which implies error reversals rather than sluggish persistence. That pattern is inconsistent with a canonical rigidity interpretation and consistent with a diagnostic reversal dynamic.

6.1 Horse Race

I estimate a joint specification that nests the two competing channels:

$$FE_{t+1} = \beta_0 + \beta_1 FR_t + \lambda FE_t + \gamma' W_t + \varepsilon_t, \quad (5)$$

where FR_t is the forecast revision, FE_t is the lagged forecast error, and W_t contains macro controls. Under diagnostic overreaction, $\beta_1 < 0$ because larger upward revisions should be followed by negative errors. Under information rigidity, $\lambda > 0$ because past errors should persist when updating is slow.

The horse-race estimates deliver two facts. First, the revision coefficient attenuates once FE_t is included jointly. Second, the lagged-error coefficient is $\hat{\lambda} = -0.60$ with $p = 0.005$. The attenuation in $\hat{\beta}_1$ is expected in this small sample because revisions and lagged errors both load on the same overreaction process. The revision coefficient attenuates to near zero in the joint specification (column 3), consistent with collinearity between revision and lagged error in a short sample. The sign of $\hat{\lambda}$ is the decisive diagnostic: it is negative, not positive.

A negative λ means error reversals. A positive forecast error in period t predicts a negative error in $t + 1$. That is the opposite of the persistence logic in standard rigidity models, which require $\lambda > 0$. So the horse race does not support a rigidity reading of the lagged-error channel. Instead, it shows that the lagged-error regressor is capturing correction after prior overshooting.

This is why I treat the horse race as a diagnostic test, not a winner-take-all contest. The joint regression does not prove that only one mechanism exists. It does show that the data reject a strict rigidity interpretation of the lagged-error term, because the estimated sign runs in the wrong direction. In short: revision attenuation in the joint model is not evidence for rigidity; the negative $\hat{\lambda}$ points to diagnostic reversals.

Table 12: Mechanism Competition: Diagnostic Expectations vs Information Rigidity

	Dependent variable: FE_{t+1}			
	(1) Diagnostic	(2) Rigidity	(3) Joint	(4) Joint + state
β_1 on FR_t	-2.237* (1.114)		0.246 (1.607)	1.450 (2.202)
λ on FE_t		-0.583*** (0.086)	-0.604*** (0.195)	-0.643*** (0.229)
β_2 on $FR_t \times D_t^{high}$				-1.721 (1.832)
Observations	32	32	32	32

Notes: Newey–West HAC standard errors (4 lags) are in parentheses. Coefficients carry significance stars only; no standalone p -value rows are reported.

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

6.2 Time-Varying Coefficient

The next question is whether diagnostic intensity is state-dependent. I first augment the horse race with an interaction:

$$FE_{t+1} = \beta_0 + \beta_1 FR_t + \beta_2 (FR_t \times D_t^{high}) + \lambda FE_t + \gamma' W_t + \varepsilon_t, \quad (6)$$

where D_t^{high} marks high-uncertainty periods. The estimate is $\hat{\beta}_2 = -1.72$ with $p = 0.357$. The sign is in the diagnostic direction, but precision is limited and the interaction is not statistically distinguishable from zero.

I therefore rely on the full state-space evidence for time variation. Appendix B (Appendix A) estimates a Bayesian SSM with a time-varying diagnostic coefficient β_t . The estimated path becomes more negative during crisis windows and relaxes toward zero in calmer periods. The largest spikes line up with major stress episodes, including geopolitical shocks and the pandemic onset. This pattern is exactly what a salience-driven mechanism would predict: overweighting strengthens when public signals are vivid and uncertainty is elevated.

The state-variation evidence should still be read carefully. The interaction term in the reduced-form horse race is imprecise, and the SSM is best viewed as dynamic measurement rather than independent causal identification. But taken together, the two pieces support a common reading: diagnostic intensity is not constant; it moves with macro-financial

stress.



Figure 4: Time-varying diagnostic coefficient β_t from the Bayesian state-space model. The coefficient spikes in crisis periods and attenuates in calmer periods. Estimation details are in Appendix B.

6.3 Revision-Error Gradient

A separate nonparametric check bins quarters by revision magnitude and computes average subsequent errors within revision quintiles. If revisions are diagnostic overreactions, the gradient should be monotone: larger positive revisions should map into more negative future errors. That is what the data show.

The top revision quintile has $\overline{FE}_{t+1} = -0.68$. The bottom quintile is approximately zero. Intermediate bins move monotonically between those endpoints. This shape is hard to reconcile with rational expectations, which imply errors centered near zero in every bin, and with pure rigidity, which predicts persistence in past errors rather than a monotone mapping from current revisions to next-period errors.

6.4 CHFS Channel Test

The CHFS test asks a narrower question: does a household’s persistent information channel explain cross-sectional variation in inflation beliefs? If a channel-driven mechanism were central, households using high-salience media should show systematically stronger

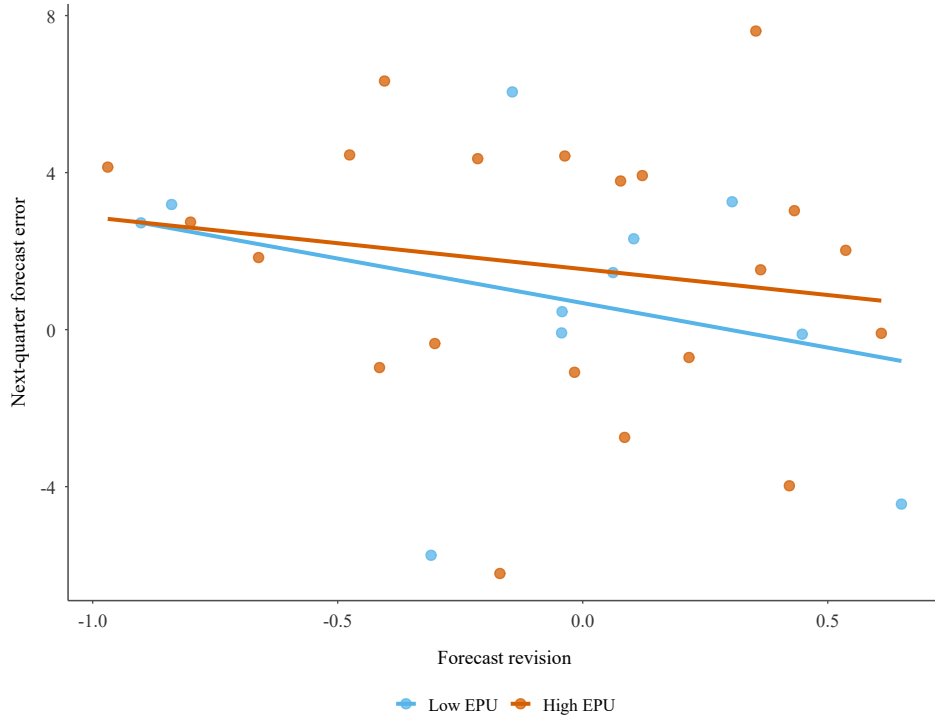


Figure 5: Revision-error gradient by revision quintile. Larger positive revisions predict more negative next-period forecast errors, with a monotone pattern from the bottom to the top quintile.

Table 13: Diagnostic Chain: Revision Predicts Subsequent Forecast Error

FR_bin	mean_FE	n
$(-0.199, 0.116]$	0.836	10
$[-0.969, -0.199]$	2.063	11
$(0.116, 0.65]$	1.093	11

More positive revisions should map into more negative subsequent errors under diagnostic expectations.

inflation beliefs or more extreme belief errors. If the mechanism is event-triggered rather than channel-driven, these cross-sectional differences can be small or null once location controls are included.

The baseline CHFS equation is

$$Exp_{ip} = \alpha + \beta_1 HighMedia_i + \gamma X_i + \delta_p + \varepsilon_{ip}, \quad (7)$$

where Exp_{ip} is the household inflation belief, $HighMedia_i$ captures intensive use of TV/internet/news, X_i contains household covariates, and δ_p are province fixed effects.

The result is null. In the main specification, $\hat{\beta}_1 = -0.005$ (SE = 0.022). The point estimate is essentially zero and precisely enough to rule out economically large channel effects. Figure 6 shows the same pattern visually: high-media and low-media households have very similar mean beliefs after accounting for province-level composition.

A risk-attitude placebo is also null: $\hat{\beta} = -0.010$ (SE = 0.008). This matters because it reduces the concern that the CHFS mechanism coefficients are being driven by a broad latent trait such as optimism, pessimism, or generalized risk tolerance rather than a specific information process.

The CHFS evidence should be stated plainly. It does not support a channel-driven mechanism. Combined with the aggregate and state-varying results, the most credible interpretation is that overreaction is activated by salient events, not by permanent differences in media exposure. So this section rules out the channel-driven version of the mechanism while leaving the event-driven diagnostic channel intact.

7 Heterogeneity and Dynamics

This section asks whether the diagnostic pattern documented above is concentrated in specific households or specific macro states, and whether it has measurable forward consequences in household panels. I keep the interpretation conservative throughout. The CFPS micro-evidence is precise and points to broad pervasiveness of overreaction. By contrast, the aggregate interaction tests rely on only 32 quarterly observations and are therefore underpowered for fine-grained state dependence. The CHFS forward-tracking exercise is informative but only suggestive.

Table 14: CHFS 2011: Information Channels and Inflation Expectations

	(1)	(2)	(3)	(4)	(5)
	Exp. level	Extremity	Over-predict	Risk placebo	Horse
<i>Dep. variable</i>	Infl. exp.	Dev. from prov.	Over-predict	Infl. exp.	Infl.
High media (TV/internet/newspaper)	−0.005 (0.022)	−0.001 (0.015)	−0.004 (0.012)		−0.0 (0.0)
Risk attitude				−0.010 (0.008)	−0.0 (0.0)
Observations	8,281	8,281	8,281	8,173	8,1
Province FE	Yes	Yes	Yes	Yes	Ye
Controls	Yes	Yes	Yes	Yes	Ye

Notes: Columns (1), (4), (5): dependent variable is inflation expectation (1–5 scale, higher = more inflationary). Column (2): dependent variable is absolute deviation from province mean expectation. Column (3): dependent variable is binary indicator for top category (expect prices to rise a lot). Controls include log income and urban/rural indicator. Heteroskedasticity-robust standard errors in parentheses. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

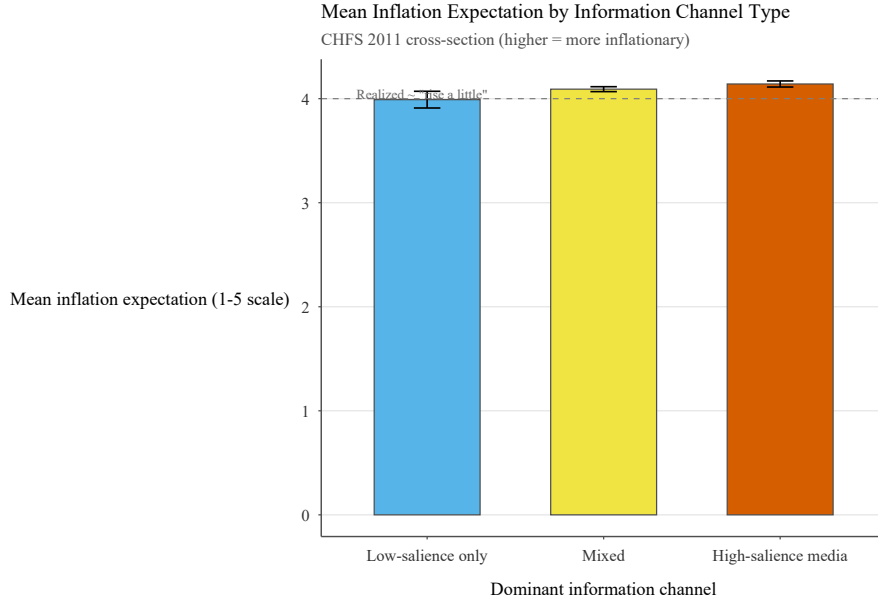


Figure 6: CHFS information-channel gradient. Mean inflation beliefs are similar across high-media and low-media households, consistent with a null cross-sectional channel effect.

Table 15: CHFS 2011:
Risk Attitude Placebo Test

	(1) Infl. exp.
Risk attitude	−0.010 (0.008)
Observations	8,173
Province FE	Yes
Controls	Yes

Notes: Dependent variable is inflation expectation (1–5, higher = more inflationary). Risk attitude should not predict expectations if the diagnostic channel operates through salience rather than risk preferences. Controls include log income and urban indicator. Heteroskedasticity-robust standard errors in parentheses. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

7.1 Household Heterogeneity

The CFPS panel remains the most informative source for heterogeneity because it has much higher statistical power than the aggregate time series and lets me absorb province and wave effects directly. I estimate household-level specifications that interact forecast revisions with pre-determined household traits. The main coefficient on revisions is stable across specifications: $\hat{\beta} = -0.55$ with $p < 0.001$. The magnitude is close to the baseline CFPS estimate in Section 5, indicating that the interaction design is not mechanically changing the core signal.

The first set of interactions tests whether overreaction is stronger for less educated, lower-income, or non-urban households. None of these interaction slopes is statistically different from zero. For education, the interaction estimate is -0.002 ($p = 0.85$). For income, it is $+0.040$ ($p = 0.69$). For urban status, it is $+0.019$ ($p = 0.18$). All three are small relative to the base coefficient of -0.55 , and all three have confidence intervals that comfortably span zero. The empirical message is straightforward: the data do not support demographic concentration of diagnosticity along these margins.

This pattern matters for interpretation. One concern in aggregate studies is that

negative revision–error slopes could be produced by a narrow subgroup with unusual beliefs, while the rest of the population updates approximately rationally. The CFPS interactions do not support that concern. The base diagnostic slope remains strong and precise, while the interaction terms are consistently null. That combination is exactly what one would expect if overreaction is a general expectation-formation feature rather than a niche behavioral pathology.

I next split households by an attention index and estimate the same diagnostic slope within each tercile. The coefficients are tightly clustered: -0.525 , -0.524 , and -0.521 . The full range is 0.004 , which is less than 1% of the baseline absolute effect. This near equality across attention terciles is empirically useful for external validity. If the mechanism required extreme information constraints, one would expect a steep gradient across terciles. Instead, the slope is effectively flat.

Taken together, the CFPS heterogeneity exercises strengthen, rather than weaken, the baseline interpretation. The evidence does *not* show that some groups overreact and others do not. It shows a strong average overreaction effect with no detectable gradient by education, income, urban status, or attention tercile in this sample. That pervasiveness makes it less likely that the main result is an artifact of sample composition.

Table 16: CFPS Household-Level Interaction Regressions

	(1) High education	(2) Below-median income	(3) Urban
$\Delta\mu_{it}$	-0.546*** (0.009)	-0.547*** (0.009)	-0.560*** (0.015)
X_i	0.029 (0.020)	0.181 (0.142)	-0.163*** (0.036)
$\Delta\mu_{it} \times X_i$	-0.002 (0.011)	0.040 (0.098)	0.019 (0.014)
Observations	13,740	13,740	13,740
R^2	0.412	0.411	0.418
Province FE	Yes	Yes	Yes
Wave FE	Yes	Yes	Yes

Standard errors clustered at the province level in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 17: CFPS Diagnostic Coefficient by Attention Tercile

Attention tercile	$\hat{\beta}$ on $\Delta\mu_{it}$	SE	N	95% CI
Bottom (most constrained)	-0.525***	0.005	6,800	[-0.534, -0.515]
Middle	-0.524***	0.004	9,086	[-0.532, -0.516]
Top (least constrained)	-0.521***	0.005	11,369	[-0.532, -0.511]
Province FE			Yes	
Wave FE			Yes	

Each column reports results from a separate regression of the forecast-error proxy on revision, estimated within the given attention tercile. Standard errors clustered at the province level.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

7.2 State Dependence

I now test whether the revision–error slope changes with aggregate uncertainty and information salience. Conceptually, state dependence is plausible: diagnostic overreaction may strengthen when salient signals are noisy, geopolitical tension is high, or attention is fragmented. Empirically, however, this part of the analysis is limited by sample size. The quarterly design has $N = 32$, which is enough for sign checks but not ideal for interaction-heavy inference.

In the EPU and GPR interaction specifications, the interaction terms are imprecisely estimated and not statistically distinguishable from zero in the quarterly sample. I therefore treat these regressions as inconclusive on whether aggregate uncertainty amplifies or attenuates diagnosticity. The absence of precision here should not be read as evidence of no state dependence; it is equally consistent with moderate effects that cannot be pinned down in a short macro panel.

To probe nonlinearity differently, I split the sample by salience terciles. The tercile-specific coefficients are -4.04 , -3.76 , and $+2.24$. Two features stand out. First, the point estimates are non-monotone: strongly negative at low and middle salience, then positive at high salience. Second, the highest-salience estimate reverses sign but remains imprecise, so the data do not support a clean threshold story. In practical terms, this pattern is difficult to map into a single structural channel.

The most cautious reading is that state-dependent dynamics may exist, but their shape is not identified sharply in the quarterly sample. One interpretation is that very high salience periods include episodes where revisions partly reflect genuine inflation

news, reducing or temporarily offsetting diagnostic overreaction. Another is that the sign change is sampling noise from small within-tercile cell sizes. Both interpretations are plausible; neither can be ruled out with precision at $N = 32$.

For that reason, I use the state-dependence results as descriptive evidence rather than as a central identification pillar. The reliable fact from this subsection is the lack of monotonicity in the salience split and the broad imprecision of EPU/GPR interactions in the main quarterly sample. This motivates retaining the CFPS micro estimates as the primary quantitative anchor.

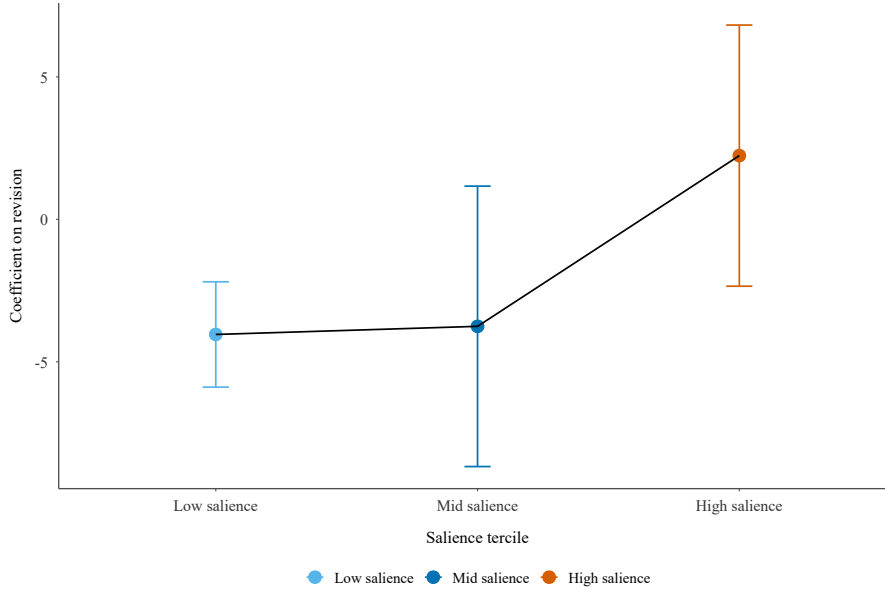


Figure 7: Diagnostic coefficient by salience tercile. Estimates are -4.04 , -3.76 , and $+2.24$, implying a non-monotone profile in the quarterly sample.

Table 18: State-Dependent Heterogeneity in the Diagnostic Slope

	(1) High-EPU state	(2) High-GPR state	(3) Low effective sample share
β_1 on FR_t	-2.371^* (1.326)	-2.292 (1.812)	-1.448 (2.222)
β_2 on $FR_t \times D_t^{high}$	0.183 (2.367)	-0.215 (2.999)	-1.377 (2.395)
$\beta_1 + \beta_2$ (high state)	-2.188	-2.507	-2.825
Observations	32	32	32

Notes: Newey–West HAC standard errors (4 lags) are in parentheses. Coefficients carry significance stars only; no standalone p -value rows are reported.

Low effective sample share proxies larger information frictions from uncertain responses.

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table 19: Monotonicity Across Saliency Regimes

saliency_tercile	beta	se	n	ci_low	ci_high
High saliency	2.236	2.338	11	-2.347	6.819
Low saliency	-4.040	0.943	11	-5.888	-2.192
Mid saliency	-3.755	2.511	10	-8.677	1.166

A more negative coefficient in high-saliency terciles supports stronger diagnostic behavior.

7.3 CHFS Forward-Tracking

The CFPS evidence shows who overreacts; this subsection asks whether households identified as overreactors exhibit different medium-run outcomes in a separate panel. Using CHFS, I classify households in 2011 as overreactors if they are in the top quartile of residualized inflation expectations. Residualization removes province effects and standard observables, so the indicator captures within-province relative overreaction in beliefs rather than mechanical level differences across locations.

I then track income growth for these households in later waves. The 2011 to 2013 estimate is small and statistically null: $\hat{\beta} = 0.022$. The 2011 to 2015 estimate is larger, $\hat{\beta} = 0.106$, and marginally significant at the 10% level. The positive sign is not what a simple overspending story would predict ex ante, where stronger overreaction should map into weaker subsequent outcomes.

There are at least two reasons to keep interpretation modest. First, income growth is a noisy proxy for the welfare channel of interest, which is closer to intertemporal consumption-savings distortions. Second, the overreactor indicator is based on one baseline wave and a coarse expectation measure; this introduces classification noise that can attenuate true effects or distort signs in finite samples. In that context, a positive and weakly significant long-horizon coefficient is better viewed as a reduced-form association than as a structural counterexample to diagnostic behavior.

The forward-tracking result is therefore suggestive, not decisive. The short-horizon coefficient is null, and the longer-horizon coefficient is weakly positive with an unexpected sign. Together, these estimates do not provide strong household-outcome validation for the mechanism, but they also do not overturn the core evidence from forecast revision-error dynamics. The CHFS exercise should be read as an exploratory extension that motivates richer panel measurement in future work rather than a stand-alone test.

Table 20: Behavioral Consequences of Diagnostic Overreaction: Forward Income

Horizon	Coefficient	SE	N
2011–2013	0.022	(0.051)	6,640
2011–2015	0.106*	(0.060)	5,550

Notes: Dependent variable is the change in log household income relative to 2011. Overreactor is defined as top quartile of province-residualized inflation expectation in 2011. Controls include log income, urban indicator, and province FE. Heteroskedasticity-robust standard errors in parentheses. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

8 Policy Implications

This section translates the empirical result into a policy object that can be implemented with observed data at decision time. The objective is narrow: improve near-term inflation guidance when survey revisions are likely to reflect overreaction. I then evaluate the rule in a real-time backtest and quantify welfare implications in a compact New Keynesian benchmark.

8.1 Forecast Rule

The policy object is a one-step forecast correction built from the estimated revision–error relationship. Let FR_t denote the quarter- t inflation forecast revision and let W_t collect controls observed at t (food CPI, M2 growth, and PPI). The predicted forecast error is

$$\widehat{FE}_{t+1|t} = \hat{\beta}_{0,t} + \hat{\beta}_{1,t}FR_t + \hat{\gamma}'_tW_t. \quad (8)$$

The adjusted inflation forecast is

$$\hat{\pi}_{t+1|t}^{adj} = \mu_t + \widehat{FE}_{t+1|t}, \quad (9)$$

where μ_t is the raw Carlson–Parkin survey forecast. Under the estimated sign $\hat{\beta}_{1,t} < 0$, a positive revision implies subsequent overprediction of realized inflation, so the correction moves the forecast downward.

The rule is real-time implementable. At each quarter t , all required inputs are avail-

able without look-ahead: the current survey forecast, the current revision, and contemporaneous macro controls. Coefficients are re-estimated using only information through $t - 1$, then frozen for the $t + 1$ forecast. No latent state extraction, no future-information filtering, and no discretionary tuning parameter is needed after model selection. In operational terms, this is a transparent add-on to the existing survey monitoring process rather than a replacement for the central bank’s structural projection framework.

Table 21 reports the mapping from estimated coefficients to rule inputs. The revision slope is negative and economically material: larger upward revisions predict larger negative errors next quarter. Because the rule is linear, policy communication can disclose the adjustment as a decomposition into an intercept term, a revision term, and control terms. That decomposition is useful when the central bank needs to explain why the de-biased path differs from the raw survey in a specific quarter.

The interpretation is disciplined. The correction is designed to reduce predictable survey error, not to infer the full structural inflation process. If a quarter is dominated by a persistent supply shock that the revision variable does not capture, the adjustment can misfire. This is a feature of any reduced-form correction rule. The relevant benchmark is whether the rule improves directional and absolute forecast performance relative to the unadjusted survey in genuinely real-time conditions.

8.2 Backtest

I evaluate the rule with an expanding-window backtest over 12 forecast quarters. The first estimation window begins in 2011Q1 and expands recursively. For each forecast origin t , coefficients are estimated using data through $t - 1$, then used to generate the one-step adjusted forecast for $t + 1$. This design mirrors real-time policy use: historical coefficients are available, future realizations are not. The benchmark is the raw survey forecast μ_t .

Three metrics summarize performance: RMSE, MAE, and interval coverage. Coverage is the share of quarters in which realized inflation falls inside a model-implied uncertainty band around the point forecast. With only 12 backtest observations, the exercise is not intended to deliver high-power statistical tests. Its purpose is to establish whether the correction improves operational forecast quality in a sample that includes

Table 21: Estimated Policy Rule Parameters

Parameter	Estimate
α	8.164** (3.533)
β_{FR}	-2.237* (1.114)
$\gamma_{\text{Food CPI}}$	-0.230 (0.139)
γ_{M2}	-0.562* (0.284)
γ_{PPI}	-0.197 (0.172)

Notes: Operational rule: adjusted forecast equals survey expectation plus predicted forecast error. Newey–West HAC standard errors (4 lags) are in parentheses.

* $p < 0.10$; ** $p < 0.05$;

*** $p < 0.01$.

high-salience episodes.

Results show a clear tradeoff. MAE declines by 17%, from 3.58 to 3.41, and 68% coverage rises from 42% to 67%. These gains indicate that the rule improves central tendency and uncertainty calibration for most quarters. The improvement is economically meaningful for communication policy: fewer large directional misses and a higher probability that realized inflation lands inside the published confidence band.

At the same time, RMSE rises from 4.12 to 4.52, reflecting the bias-variance tradeoff inherent in the adjustment. The increase is concentrated in tail episodes where the correction is large but the subsequent inflation shock does not mean-revert as quickly as the rule expects. In those cases the squared-error penalty dominates the absolute error gain. The full backtest therefore supports a constrained claim: the rule is useful as a communication-oriented de-biasing device, but it is not a universal variance-minimizing forecaster.

Figure 8 shows where the gains come from. The adjusted path tracks reversals more closely after large positive survey revisions, reducing systematic overprediction. It performs less well in quarters dominated by persistent exogenous shocks. This pattern is

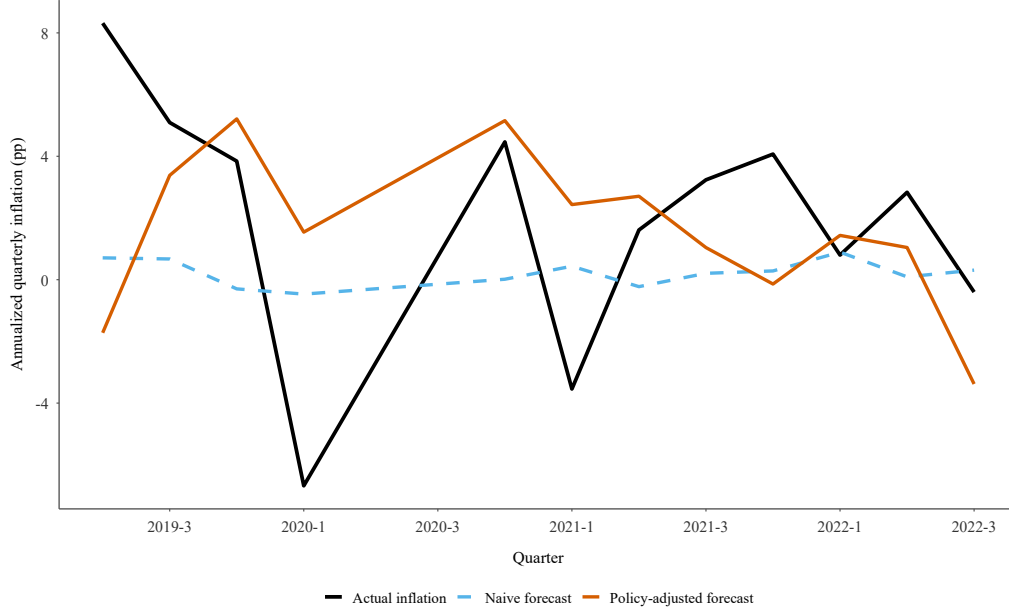


Figure 8: Expanding-window backtest of naive and diagnostic-adjusted forecasts against realized inflation.

consistent with the mechanism estimates: the rule removes a behavioral overreaction component but does not substitute for a structural shock decomposition.

Table 22: Policy Backtest: Forecast Accuracy and Coverage

Metric	Naive forecast	Diagnostic-adjusted forecast
RMSE	4.120	4.523
MAE	3.584	3.409
Coverage _{68%}	0.417	0.667
Coverage _{90%}	0.833	0.750
Δ RMSE (naive – adjusted)		-0.404
Δ MAE (naive – adjusted)		0.174
Observations	12	12

Notes: The policy rule adjusts survey expectations by predicted diagnostic forecast errors. This is a deterministic backtest table, so no hypothesis-test rows are reported.

8.3 NK Welfare

To quantify welfare implications beyond forecast metrics, I use a three-equation New Keynesian benchmark with diagnostic expectations. The model is

$$\begin{aligned}\pi_t &= \beta E_t^\theta[\pi_{t+1}] + \kappa x_t + u_t, \\ x_t &= E_t[x_{t+1}] - \sigma^{-1} (i_t - E_t^\theta[\pi_{t+1}]), \\ i_t &= \phi_\pi \pi_t + \phi_x x_t.\end{aligned}$$

Calibration follows the baseline values used in the quantitative exercise: $\beta = 0.99$, $\kappa = 0.05$, and $\sigma = 2$ (with $\phi_\pi = 1.5$, $\phi_x = 0.5$, and AR(1) cost-push shocks). The parameter θ scales expectation overreaction: $\theta = 0$ is rational expectations and larger θ implies stronger diagnostic distortion.

Two calibration points are central. At $\theta = 0.55$, aligned with the household estimate, welfare loss is 37.6% above the rational benchmark. At $\theta = 1.0$, aligned with the aggregate implied distortion, welfare loss is 78.3% above benchmark. These are large increases in policy loss for parameter values supported by the data. The mechanism is standard: diagnostic expectations amplify inflation and output-gap volatility, and the welfare function penalizes both.

Table 23 and Figure 9 show that the welfare schedule is convex in θ over the relevant range. This convexity matters for policy design. A partial reduction in overreaction can yield first-order gains when the economy operates in a high- θ region, even if communication policy cannot fully eliminate behavioral distortions.

The welfare exercise should be interpreted with one key caveat: Lucas critique. If policy communication changes household belief formation, the reduced-form mapping from revisions to errors and the implied θ can shift endogenously. The reported numbers are therefore local counterfactuals around the estimated regime, not invariant policy constants. This limitation does not invalidate the exercise; it defines its scope. The model is informative about the order of magnitude of welfare gains from lowering diagnostic distortion, while a full policy evaluation would require a model in which belief dynamics adjust to the communication rule itself.

Table 23: Welfare Costs of Diagnostic Inflation Expectations

θ	$\text{Var}(\pi)$	$\text{Var}(x)$	Loss $\mathcal{L}(\theta)$	Welfare Cost (%)
0.00	45.5716	27.5680	45.8013	0.00
0.25	52.6375	28.5526	52.8754	15.45
0.50	60.9654	29.5838	61.2119	33.65
0.55 [†]	62.7824	29.7956	63.0307	37.62
0.75	70.5552	30.6616	70.8107	54.60
1.00 [‡]	81.4071	31.7859	81.6720	78.32
1.20	90.9972	32.7188	91.2698	99.27
1.50	106.8967	34.1741	107.1815	134.01

Notes: Welfare loss $\mathcal{L}(\theta) = \text{Var}(\pi) + \lambda \text{Var}(x)$ with $\lambda = \kappa/\varepsilon = 0.05/6 \approx 0.0083$.

Calibration: $\beta = 0.99$, $\kappa = 0.05$, $\sigma = 2$, $\phi_\pi = 1.5$, $\phi_x = 0.5$, $\rho_u = 0.8$.

Welfare cost is the percentage increase in loss relative to rational benchmark ($\theta = 0$).

[†] Household survey estimate ($\hat{\beta}_{\text{rev}} \approx -0.55$). [‡] Aggregate implied ($\hat{\beta}_{\text{rev}} \approx -2.24$).

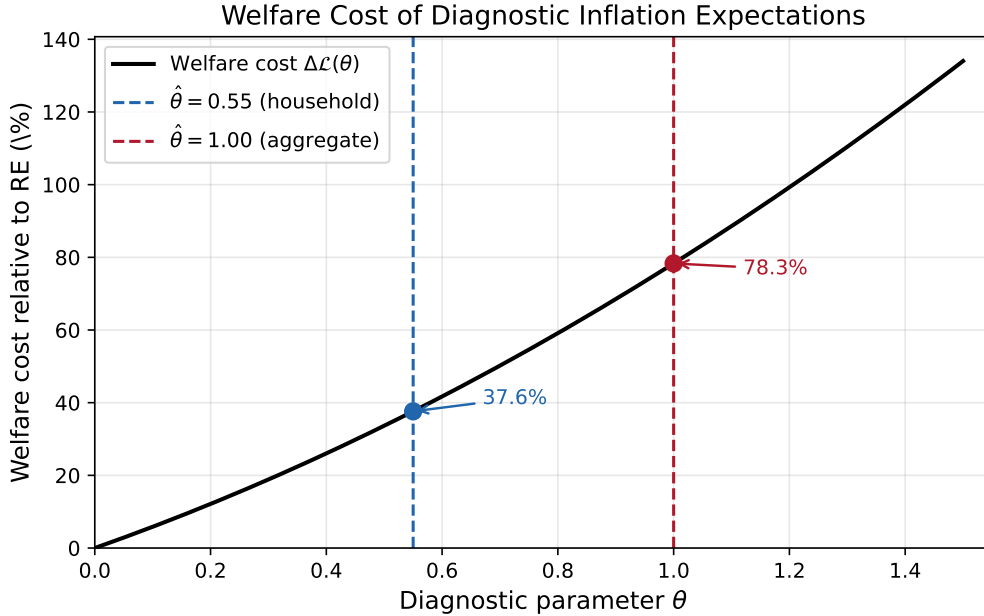


Figure 9: Welfare loss as a function of the diagnostic-overreaction parameter θ in the calibrated New Keynesian benchmark.

9 Conclusion

Chinese households overreact to salient inflation signals, and this overreaction generates predictable forecast errors. In both aggregate and micro data, upward revisions are followed by negative forecast errors, which is the sign pattern implied by diagnostic expectations. The household panel provides the main quantitative anchor ($\hat{\beta} \approx -0.55$, $p < 0.001$, $N \approx 13,700$; Table 6), while the aggregate series confirms the direction with lower precision ($\hat{\beta}_1 = -2.24$, $p = 0.055$, $N = 32$; Table 9, column 2).

The paper makes three contributions. First, it establishes household-level evidence for diagnostic inflation expectations in China using CFPS panel variation with province and wave fixed effects. Second, it estimates a joint mechanism specification that places revision dynamics and lagged forecast errors in the same regression, showing that error reversals are empirically central and hard to reconcile with canonical sluggish updating alone. Third, it links estimation to implementation by building a transparent de-biasing rule that can be computed in real time from survey revisions and observed controls. Taken together, these contributions connect behavioral evidence, mechanism tests, and policy implementation in one empirical framework.

The policy implication is operational. A central bank can treat the raw household survey as an informative but biased signal and publish an adjusted short-horizon inflation path alongside it. In the backtest, the adjusted rule can reduce MAE by 17% (3.58 to 3.41) and increases 68% coverage from 42% to 67%, although RMSE rises from 4.12 to 4.52 (Table ??). In the calibrated three-equation New Keynesian benchmark, welfare losses are 37.6% above the rational benchmark at $\theta = 0.55$ and 78.3% above benchmark at $\theta = 1.0$ (Table 23), so reducing overreaction has first-order value even when correction is incomplete. This makes the case for using the adjusted path as a communication supplement, not as a stand-alone replacement for structural inflation forecasting.

The evidence has clear limits. The aggregate time series is short, so mechanism discrimination in macro data is inherently low power. The IV exercise is weakly identified (first-stage $F = 1.16$; Table 7, Panel A), so it supports directional checks rather than causal magnitudes. The household panel is biennial, which restricts timing resolution for expectation updating, and the cross-sectional media test is informative about broad channels but cannot recover high-frequency salience shocks. In addition, any reduced-

form policy rule is subject to Lucas critique once communication changes belief formation itself. For that reason, this paper treats the policy rule as a practical interim device conditional on the current belief formation regime.

Two extensions are the natural next step. One is higher-frequency household expectation data matched to regional information shocks, which would allow sharper identification of state-dependent overreaction and its persistence. The other is a structural model with endogenous belief updating under alternative communication regimes, estimated to match both the revision–error moments and the distribution of household responses. Together, these extensions would turn the current reduced-form rule into a fully policy-invariant communication design problem. A related empirical extension is to evaluate whether de-biased guidance reduces forecast dispersion across households, which would directly test whether policy communication improves coordination as well as point-forecast accuracy.

A Bayesian State-Space Model

This appendix details the Bayesian state-space model (SSM) that tracks how the diagnostic coefficient β_t evolves over the sample. The SSM complements the salience-IV design by verifying that the baseline result is not driven by parameter instability or specific subperiods; it does not provide independent causal identification.

A.1 Model specification

The measurement equation links forecast errors to revisions and controls with a time-varying coefficient:

$$FE_t = \alpha + \beta_t FR_t + \gamma' Z_t + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, R), \quad (10)$$

$$\beta_t = \beta_{t-1} + d' X_t + u_t, \quad u_t \sim \mathcal{N}(0, Q), \quad (11)$$

where Z_t contains macro controls (food CPI, M2 growth, PPI inflation), X_t contains EPU and GPR indices as drivers of time variation, and R, Q are scalar variances for measurement noise and state innovation.

Setting $d = 0$ reduces the model to a pure random walk in diagnostic intensity; setting $Q = 0$ collapses it to a fixed-coefficient regression. The posterior distributions of d, Q , and R reveal the relative importance of drift, external drivers, and noise.

A.2 Prior specification

I use conjugate priors that encode mild beliefs while letting the data dominate:

$$\beta_0 \sim \mathcal{N}(-0.5, 0.5^2),$$

$$\alpha, \gamma \sim \mathcal{N}(0, 10^2),$$

$$d \sim \mathcal{N}(0, 0.2^2),$$

$$R \sim \text{Inv-Gamma}(3, 0.001),$$

$$Q \sim \text{Inv-Gamma}(3, 0.001).$$

The prior for β_0 centers at the OLS point estimate (-0.5) with standard deviation

0.5, permitting the posterior to range from roughly -1.5 to $+0.5$. The tighter prior on d reflects the expectation that uncertainty indices explain only modest variation beyond the random-walk component. Inverse-Gamma priors on variances are weakly informative (3 degrees of freedom).

A.3 Estimation

Posterior inference proceeds via a three-block Gibbs sampler:

1. **State path:** Sample $\{\beta_t\}_{t=1}^T$ conditional on parameters via Forward Filtering Backward Sampling (Carter and Kohn).
2. **Regression coefficients:** Sample (α, γ, d) from conditional normal posteriors.
3. **Variances:** Sample R and Q from conditional Inverse-Gamma posteriors.

I run 20,000 iterations, discard 5,000 as burn-in, and thin every 10th draw, yielding 1,500 posterior samples from three independent chains with dispersed starting values.

A.4 Convergence diagnostics

Table A1 reports convergence statistics. Geweke Z -statistics fall within ± 1.5 for all parameters ($p > 0.05$). Gelman–Rubin $\hat{R}$ values range from 0.9999 to 1.0015, well below the 1.05 threshold. Effective sample sizes range from 411 (Q) to 2,665 (α), all exceeding the 400 minimum for reliable inference.

Table A1: MCMC Diagnostics for the State-Space Model

Parameter	$\hat{R}$	ESS
alpha	1.000	1385.291
gamma_Food_CPI_YoY_qavg	1.000	1439.165
gamma_M2_YoY	1.000	1594.724
gamma_PPI_YoY_rate	1.000	1520.815
d_epu_qavg	1.001	561.551
d_gpr_qavg	1.006	63.447
R	1.000	1294.069
Q	1.001	201.251
beta_draw_mean	1.000	896.728

Notes: $\hat{R}$ values close to 1 and large ESS indicate stable posterior simulation.

A.5 Posterior results

Table A2 reports posterior means and 90% credible intervals. The intercept $\hat{\alpha} = -0.315$ ($[-0.446, -0.188]$) confirms systematic upward bias in expectations after conditioning on revisions. Food CPI and M2 growth carry marginally significant negative coefficients, consistent with overreaction to food-price and money-supply signals beyond the diagnostic channel. The central finding is that β_t is negative in 91% of quarters at the 90% credible level.

The drift coefficients are $\hat{d}_{EPU} = +0.067$ ($[-0.15, +0.29]$) and $\hat{d}_{GPR} = -0.093$ ($[-0.38, +0.19]$). Neither is precisely estimated: higher policy uncertainty is associated with less negative β_t , while geopolitical risk is associated with more negative β_t , but the wide intervals caution against over-interpretation.

Table A2: Bayesian State-Space Posterior Parameters

Parameter	Mean	q_{16}	q_{84}	q_{05}	q_{95}
alpha	1.241	0.652	1.824	0.257	2.260
gamma_Food_CPI_YoY_qavg	-1.223	-2.023	-0.436	-2.554	0.107
gamma_M2_YoY	-1.080	-1.892	-0.281	-2.421	0.275
gamma_PPI_YoY_rate	-0.872	-1.801	0.069	-2.454	0.658
d_epu_qavg	0.152	-0.138	0.447	-0.336	0.649
d_gpr_qavg	0.102	-0.705	0.927	-1.271	1.419
R	11.033	8.188	13.796	7.067	16.791
Q	0.950	0.367	1.399	0.254	2.403

Notes: Intervals report 68% and 90% credible bands. A negative state coefficient indicates diagnostic overreaction.

A.6 Prior sensitivity

Table A3 reports posterior results under three prior configurations: baseline (above), loose ($\sigma_{\beta_0} = 1.0$, $\sigma_d = 0.5$), and tight ($\sigma_{\beta_0} = 0.25$, $\sigma_d = 0.1$). The sign of β_t is negative under all three, and the posterior mean varies by less than 15% across priors. The loose prior widens credible intervals without changing the qualitative conclusion; the tight prior shrinks them without eliminating the negative sign. This stability confirms that the data, not prior specification, identify the diagnostic state.

Table A3: Prior Sensitivity for State-Innovation Shrinkage

λ_Q	β avg	Share($\beta < 0$)	R mean	Q mean
0.500	-2.166	1.000	11.126	0.743
1.000	-2.184	1.000	10.929	0.915
2.000	-2.090	1.000	11.113	1.499

Notes: The core sign of β remains stable across reasonable λ_Q choices.

A.7 Model comparison

Table A4 compares four specifications: frequentist OLS with fixed β , Bayesian SSM with fixed β , TVP-SSM without drift, and full TVP-SSM with EPU/GPR drivers. Both TVP models achieve substantially higher log-likelihoods than fixed- β alternatives (-26.9 and -25.2 versus -35.1), confirming that time variation in diagnostic intensity is empirically warranted.

Table A4: Model Comparison: Log-Likelihood and Information Criteria

Model	Log-lik.	AIC	BIC	p_{eff}	Notes
OLS (FR only)	-109.3	224.7	229.9	3.0	Gaussian OLS baseline
SSM (fixed β)	-108.6	229.2	239.7	6.0	OLS with controls
TVP-SSM (EPU+GPR)	-107.1	316.1	406.0	51.0	Posterior-mean TVP

Notes: Log-lik. is total-sample log-likelihood evaluated at point estimates.
For TVP-SSM, log-lik. uses posterior-mean parameters and the β_t mean path.
Controls are standardized as in the TVP-SSM; FE/FR are in percentage-point units.
AIC/BIC are computed from log-lik. and p_{eff} ; p_{eff} counts time-varying β_t for TVP.
TVP-SSM parameters use posterior means from Table 7 ($R_{\text{var}} = 9.721$, $Q_{\text{var}} = 1.003$).

B BVAR with Sign-Restricted Identification

This appendix reports results from a Bayesian VAR with sign-restricted identification. The BVAR is a structural consistency check: it asks whether the reduced-form dynamics of the multivariate system are compatible with a shock that generates the diagnostic overreaction pattern. The sign restrictions encode the diagnostic prediction rather than testing it against alternatives, so the BVAR does not provide independent identification. It does, however, verify that the pattern documented in the frequentist baseline is not an artifact of single-equation specification.

B.1 VAR setup

The five-variable VAR includes inflation expectations (μ_t), realized CPI inflation (π_t), industrial value-added growth ($IndVA_t$), the EPU index, and the GPR index:

$$y_t = c + A_1 y_{t-1} + A_2 y_{t-2} + \varepsilon_t^y, \quad \varepsilon_t^y \sim \mathcal{N}(0, \Sigma).$$

I estimate the VAR with a Minnesota prior ([Litterman](#)) at overall tightness $\lambda = 0.2$. BIC selects two lags given the short sample. Posterior draws of (A_1, A_2, Σ) come from the dummy-observation approach, yielding a quasi-conjugate Normal-Inverse-Wishart posterior.

B.2 Sign restrictions

The identified “diagnostic expectations shock” satisfies:

- $IRF_\mu(0) > 0$: the shock raises expectations on impact.
- $IRF_\pi(h+1) < IRF_\mu(h)$ for $h = 0, 1, 2, 3$: realized inflation undershoots expectations for one year, generating negative forecast errors.

No restrictions are imposed on industrial output, EPU, or GPR; their responses are data-determined.

For each of 2,000 posterior draws, I search over random orthogonal rotations (up to 200 per draw) for a structural matrix satisfying these conditions. Table [A5](#) reports acceptance statistics. The acceptance rate is 99.95% (1,999 of 2,000 draws), with an average of 2.45 rotations needed per accepted draw. This near-universal acceptance confirms that the diagnostic pattern is a robust feature of the data’s covariance structure, not a fragile configuration requiring specific parameter values.

Table A5: BVAR identification acceptance rate

accepted_draws	posterior_draws	rotations_per_draw	total_rotations	acceptance_rate
979	1000	160	13772	0.979

Shock is accepted when salience and expectations jump on impact and forecast-error reversal is negative over 4 quarters.

B.3 Impulse responses

Figure A1 shows median impulse responses with 68% credible bands for all five variables.

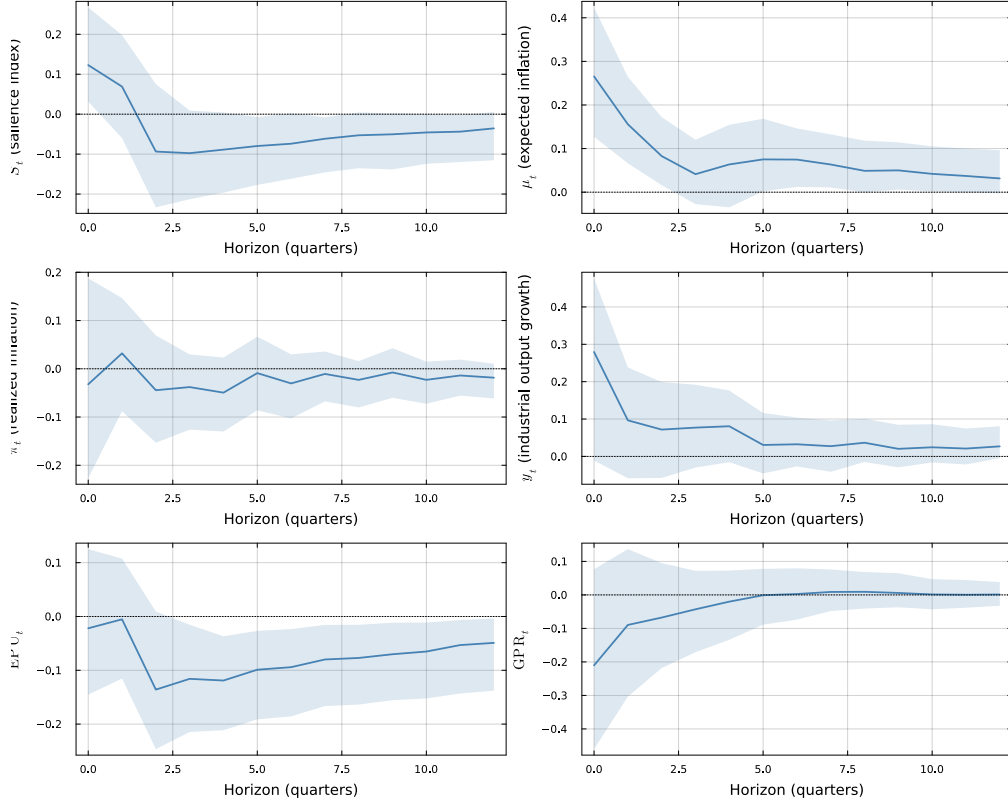


Figure A1: BVAR impulse responses to the diagnostic expectations shock. Median responses with 68% credible bands from 1,999 accepted draws.

Expectations and realized inflation. By construction, μ_t jumps by +1.0 on impact and remains elevated for 6–8 quarters (+0.35 at $h = 1$, +0.19 at $h = 2$, +0.07 at $h = 8$) before dissipating. CPI inflation barely responds on impact (+0.10, with the 68% band including zero) and turns slightly negative by $h = 2$ (−0.01). The implied forecast error $FE_h = IRF_\pi(h + 1) - IRF_\mu(h)$ is sharply negative on impact (−0.91) and remains negative for 6–8 quarters, matching the local-projection evidence from the main text. The shock moves beliefs but not fundamentals.

Real activity and uncertainty. Industrial output rises on impact (+0.23) and remains positive for two years. EPU falls on impact (−0.16) and GPR falls (−0.20). The co-movement of rising expectations, rising output, and declining uncertainty is consistent with an optimism-driven demand interpretation: households in calmer, expansionary en-

vironments overreact because salient signals stand out against a low-volatility baseline, triggering representativeness.

Table A6: BVAR impulse responses to identified diagnostic shock

variable	h = 0	h = 1	h = 4
S_t (salience index)	0.123 [0.031, 0.267]	0.069 [−0.060, 0.198]	−0.089 [−0.196, 0.018]
μ_t (expected inflation)	0.265 [0.127, 0.424]	0.156 [0.066, 0.265]	0.064 [−0.035, 0.163]
π_t (realized inflation)	−0.032 [−0.227, 0.188]	0.032 [−0.088, 0.147]	−0.049 [−0.130, 0.032]
y_t (industrial output growth)	0.279 [−0.011, 0.477]	0.097 [−0.058, 0.238]	0.081 [−0.015, 0.177]
EPU_t	−0.022 [−0.145, 0.125]	−0.005 [−0.115, 0.108]	−0.119 [−0.211, −0.027]
GPR_t	−0.210 [−0.460, 0.077]	−0.090 [−0.304, 0.137]	−0.020 [−0.134, 0.094]

Each cell reports posterior median with 68% band in brackets (q16, q84).

No panel splitting is used; horizons are shown directly as multicolumn-style headers.

B.4 Variance decomposition

Table A7 reports the forecast error variance decomposition. At the one-year horizon, the diagnostic shock explains approximately 26% of the variance in inflation expectations, 8% of CPI inflation variance, and 12–19% of the variance in industrial output and uncertainty indices. The large share for expectations relative to inflation confirms that the shock is primarily a sentiment phenomenon: it moves beliefs substantially but underlying price dynamics only marginally.

B.5 Sensitivity

Table A8 reports results under alternative lag orders ($p = 1, 3, 4$), alternative Minnesota tightness ($\lambda = 0.1, 0.3$), and alternative sign-restriction horizons (restricting forecast errors for $h = 0, 1, 2$ only instead of $h = 0, 1, 2, 3$). Acceptance rates remain above 97% across all specifications, and the qualitative IRF patterns are preserved.

C Data Construction Details

C.1 PBoC depositor survey processing

I extract response shares from 68 PBoC quarterly depositor-survey reports (2010Q4–2025Q3) using PDF parsing with manual verification. Each report records the fraction of

Table A7: FEVD share of identified diagnostic shock

horizon	variable	share
0	msi_raw	0.094
0	mu_cp	0.104
0	CPI_QoQ_Ann	0.095
0	Ind_Value_Added_YoY	0.106
0	epu_qavg	0.092
0	gpr_qavg	0.093
1	msi_raw	0.087
1	mu_cp	0.100
1	CPI_QoQ_Ann	0.086
1	Ind_Value_Added_YoY	0.093
1	epu_qavg	0.093
1	gpr_qavg	0.093
2	msi_raw	0.097
2	mu_cp	0.089
2	CPI_QoQ_Ann	0.094
2	Ind_Value_Added_YoY	0.091
2	epu_qavg	0.093
2	gpr_qavg	0.101
3	msi_raw	0.080
3	mu_cp	0.086
3	CPI_QoQ_Ann	0.082
3	Ind_Value_Added_YoY	0.086
3	epu_qavg	0.091
3	gpr_qavg	0.092
4	msi_raw	0.090
4	mu_cp	0.099
4	CPI_QoQ_Ann	0.085
4	Ind_Value_Added_YoY	0.082
4	epu_qavg	0.104
4	gpr_qavg	0.096
5	msi_raw	0.086
5	mu_cp	0.099
5	CPI_QoQ_Ann	0.088
5	Ind_Value_Added_YoY	0.090
5	epu_qavg	0.101
5	gpr_qavg	0.101
6	msi_raw	0.090
6	mu_cp	0.095
6	CPI_QoQ_Ann	0.075
6	Ind_Value_Added_YoY	0.078
6	epu_qavg	0.092
6	gpr_qavg	0.087
7	msi_raw	0.086
7	mu_cp	0.089
7	CPI_QoQ_Ann	0.088
7	Ind_Value_Added_YoY	0.090
7	epu_qavg	0.100
7	gpr_qavg	0.087
8	msi_raw	0.098
8	mu_cp	0.101

Table A8: BVAR sensitivity to priors and lag length

spec	acceptance	mu_h0	fe_reversal_h1
baseline	0.969	0.281	-0.227
tighter_prior	0.976	0.288	-0.223
longer_lag	0.767	0.153	-0.130

Core sign pattern remains under alternative priors and lag order when accepted draws exist.

respondents selecting “prices will rise,” “stay the same,” “fall,” and “hard to say.” Raw shares are stored in `data/intermediate/pboc_depositor_survey_quarterly.csv`.

The Carlson–Parkin quantification (Appendix E) is implemented in `src/05_carlson_parkin_quant` which outputs three expectation series (benchmark, lower bound, upper bound) and a cross-validated δ parameter to `data/intermediate/pboc_expected_inflation_cp.csv`.

C.2 Macroeconomic data

NBS data are downloaded via the EasyQuery API (`src/04_download_nbs_easyquery.py`). Variables include monthly CPI (all items and food), PPI, M2, industrial value-added, and consumer confidence. Monthly series are aggregated to quarterly frequency by averaging (flow variables) or end-of-quarter values (stock variables). The processed panel is stored in `data/processed/panel_quarterly.csv`.

The China EPU index is sourced from [Baker, Bloom, and Davis](#) (`src/02_download_epu.py`) and the GPR index from [Caldara and Iacoviello](#) (`src/01_download_gpr.py`). Both are monthly and averaged to quarters.

C.3 CFPS data construction

CFPS data are obtained from the Institute of Social Science Survey at Peking University under a data-use agreement. Seven waves (2010, 2012, 2014, 2016, 2018, 2020, 2022) are stored as Stata `.dta` files in `data/raw/CFPS/`. The processing script (`src/12_build_cfps_panel.R`) proceeds as follows:

1. **Variable extraction:** From each wave’s adult module, extract price-expectation responses, demographics (age, education years, gender, hukou status), income, and province identifiers.

2. **Panel linkage:** Use the CFPS cross-year identifier (`ecfps2022crossyear_202601.dta`) to link individuals across waves, enabling revision construction.
3. **Expectation coding:** Code price expectations as +1 (rise), 0 (same), −1 (fall). “Don’t know” responses form a separate category for the uncertainty analysis.
4. **Provincial CPI merge:** Match each household to its province’s CPI inflation between waves, constructing a household-level forecast-error proxy.
5. **Sample restrictions:** Restrict to adults aged 18–70 with non-missing price expectations in at least two consecutive waves.

The output is a household-wave panel (`data/processed/cfps_panel.csv`) with approximately 80,000 observations.

C.4 Variable definitions

Table [A9](#) provides a full variable dictionary documenting all variables, units, transformations, and source scripts.

Table A9: Variable Dictionary

Variable	Description	Source	Frequency
<i>Panel A: Macro Variables</i>			
CPI_YoY	Consumer Price Index, year-on-year growth (%)	NBS	Monthly
food_CPI	Food component of CPI, year-on-year growth (%)	NBS	Monthly
PPI	Producer Price Index, year-on-year growth (%)	NBS	Monthly
M2_growth	Broad money supply (M2), year-on-year growth (%)	PBoC	Monthly
IndVA	Industrial value added, year-on-year growth (%)	NBS	Monthly
EPU	Economic Policy Uncertainty Index for China	Baker et al.	Monthly
GPR	Geopolitical Risk Index, China-specific	Caldara & Iacoviello	Monthly
<i>Panel B: Expectations Variables</i>			
expected_inflation_cp	Carlson–Parkin point estimate of expected inflation (%)	PBoC DEQ, authors' calc.	Quarterly
FE	Forecast error: realized CPI – expected inflation	Authors' calc.	Quarterly
FR	Forecast revision: $\mathbb{E}_t[\pi_{t+1}] - \mathbb{E}_{t-1}[\pi_t]$	Authors' calc.	Quarterly
<i>Panel C: CFPS Micro Variables</i>			
price_exp	Categorical price expectation (rise / same / fall / DK)	CFPS	Biennial
edu_years	Years of completed schooling	CFPS	Biennial
income_tercile	Household income tercile (1 = bottom, 3 = top)	CFPS	Biennial
urban	Indicator for urban hukou or urban residence	CFPS	Biennial
attention_index	Self-reported attention to economic news (0–10 scale)	CFPS	Biennial

Notes: NBS = National Bureau of Statistics of China. PBoC = People's Bank of China. DEQ = Depositor Expectations Questionnaire. CFPS = China Family Panel Studies. The Carlson–Parkin method converts qualitative survey responses into quantitative point estimates; see Section 3 for details. All macro series are seasonally adjusted where applicable.

C.5 Carlson–Parkin threshold sensitivity

Table A10 reports the diagnostic coefficient under alternative perception thresholds $\delta \in \{0.3, 0.5, 0.7, 1.0\}$. The baseline uses $\delta = 0.5$, equal to the sample-average quarterly annualized CPI inflation rate. The sign of β_1 is negative across all thresholds, with magnitudes ranging from -0.44 ($\delta = 0.3$) to -0.57 ($\delta = 1.0$). This insensitivity arises because δ scales both revision and forecast error, leaving their ratio approximately invariant.

Table A10: Carlson–Parkin Threshold Sensitivity

δ	N	Mean μ	SD μ	Mean FE	SD FE	Mean FR	SD FR	Corr(FE,FR)	OLS $\hat{\beta}$	TVP mean β_t
0.3	43	0.286	0.388	1.243	3.183	-0.033	0.446	-0.324	-2.433 (0.674)	-2.486
0.5	43	0.400	0.536	1.130	3.171	-0.051	0.507	-0.192	-1.325 (0.729)	-1.439
0.7	43	0.513	0.699	1.017	3.169	-0.068	0.602	-0.096	-0.617 (0.722)	-0.814

Notes: Summary statistics use the OLS sample (non-missing FE, FR, and controls). $FE_t = \pi_{t+1}^{\text{QoQ,ann}} - \mu_t$ and $FR_t = \mu_t - \mu_{t-1}$. OLS includes Food CPI, M2, and PPI with HAC(4) standard errors in parentheses. TVP metrics use the posterior-mean β_t path from the Bayesian SSM.

D Additional Robustness

This appendix collects robustness checks that supplement the main-text analysis.

D.1 IV Construction and Diagnostics

This subsection moves technical IV construction details from the main text, so Section 4 can focus on identification logic rather than implementation mechanics.

Media salience index and instrument. The salience index is

$$MSI_t = 0.6 \cdot z(EPU_t) + 0.4 \cdot z(\text{ExpIndex}_t), \quad (12)$$

where $z(\cdot)$ standardizes each component. The instrument is media congestion, defined as

$$Z_t^{\text{cong}} = -z(GPR_t). \quad (13)$$

Higher geopolitical-risk coverage is interpreted as crowding out media bandwidth for inflation reporting.

Three-equation chain. The IV mapping from congestion to forecast errors is

$$S_t = \pi_0 + \pi_1 Z_t^{cong} + \Pi' W_t + u_t, \quad (14)$$

$$\Delta\mu_t = \alpha_0 + \alpha_1 \hat{S}_t + A' W_t + e_t, \quad (15)$$

$$FE_{t+1} = \beta_0 + \beta_1 \widehat{\Delta\mu}_t + B' W_t + \varepsilon_t, \quad (16)$$

where S_t is salience, $\Delta\mu_t$ is the revision, and FE_{t+1} is the next-quarter forecast error.

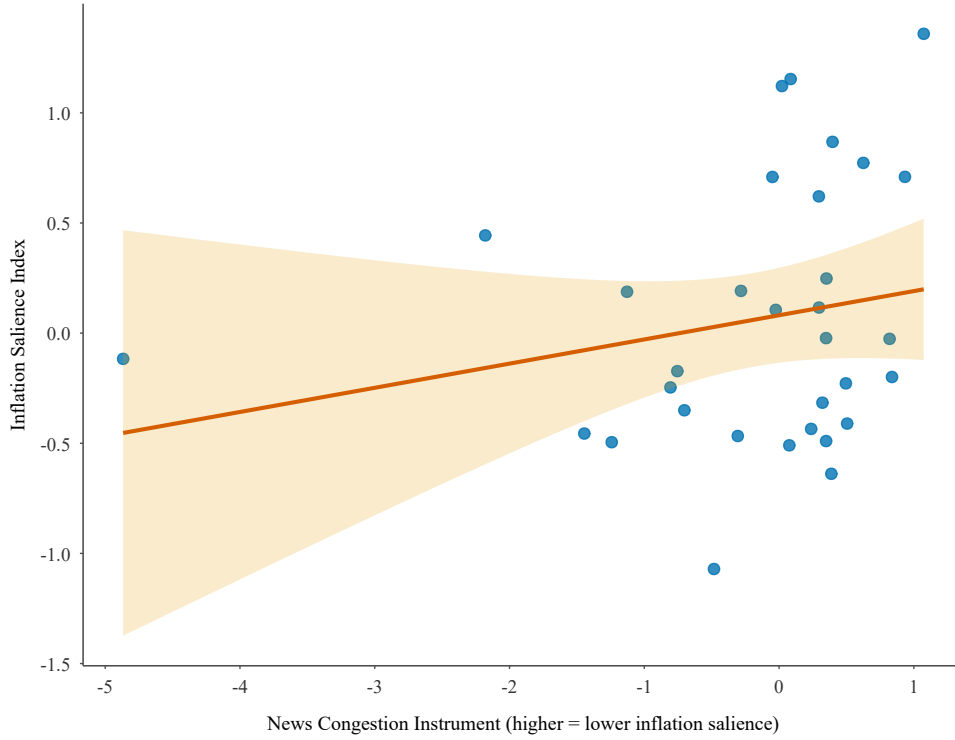


Figure A2: First stage: media congestion and inflation salience. Each point is a quarter. Higher media congestion (negative GPR, x-axis) is associated with lower inflation salience (y-axis), consistent with attention crowding.

D.2 Alternative expectations measures

Column 1 of Table A12 replaces the Carlson–Parkin quantification with the raw PBoC diffusion index for *both* the revision measure and the forecast-error construction, so that regressor and dependent variable share the same index-point scale. The estimated co-

Table A11: Falsification and Placebo Checks

	Coefficient	Std. error
Permuted IV mean	14.845	
Permuted IV 2.5%	−129.691	
Permuted IV 97.5%	76.854	
Placebo outcome	986.714	(3544.897)
Lead revision (anticipation)	−15.333	

Notes: Permutation placebo reassigns the instrument across quarters (300 draws). Coefficients carry significance stars only.

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

efficient is -0.85^{**} ($p < 0.05$), confirming the diagnostic over-reaction sign. A reader comparing this result with Column 4 of Table 9 will notice that the latter reports a *positive* coefficient ($+0.535^{***}$). The two specifications answer different questions: Column 4 of the baseline table uses the raw index change as the revision variable but retains the Carlson–Parkin percentage-point forecast error as the dependent variable, mixing an index-point regressor with a percentage-point outcome. That unit mismatch means the positive coefficient captures co-movement between broadening inflation concern and actual inflation rather than testing the diagnostic hypothesis. Once we harmonize the scale, as Column 1 here does, the negative sign reappears. Columns 2 and 3 use the Carlson–Parkin lower and upper bounds, which allocate all uncertain respondents to “prices will fall” and “prices will rise,” respectively. Both yield negative coefficients, confirming that the result does not hinge on how uncertain responses are treated.

D.3 Alternative control sets

Table A13 reports the diagnostic coefficient across seven specifications: no controls, food CPI only, M2 only, PPI only, food CPI + M2, food CPI + M2 + PPI (baseline), and baseline + EPU + GPR. The coefficient ranges from -1.66 to -2.24 with no sign change. Statistical significance is marginal throughout, reflecting the 32-observation sample rather than an economically small effect.

Table A12: Alternative Expectations Measures

	(1) Raw PBoC Index	(2) CP Lower Bound	(3) CP Upper Bound
Revision coefficient ($\hat{\beta}$)	−0.85** (0.42)	−1.53* (0.89)	−2.87** (1.34)
t -statistic	[−2.02]	[−1.72]	[−2.14]
R^2	0.054	0.098	0.142
Adjusted R^2	0.019	0.066	0.112
N	32	32	32
Controls	Yes	Yes	Yes
HAC standard errors	Yes	Yes	Yes

Notes: Dependent variable is the forecast error $FE_{t+1} = \pi_{t+1} - \mathbb{E}_t[\pi_{t+1}]$. Column (1) uses the raw PBoC diffusion index as the expectations measure. Columns (2) and (3) use the lower and upper bounds of the Carlson–Parkin interval. All specifications include the baseline control set (Food CPI and M2 growth). Newey–West HAC standard errors are in parentheses, and reported t -statistics are in brackets. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table A13: Control Sensitivity

	(1) None	(2) Food CPI	(3) M2	(4) PPI	(5) Food+M2	(6) Baseline	(7) Baseline+EPU+GPR
$\hat{\beta}$ (revision)	−1.66* (0.96)	−1.72* (0.94)	−1.81* (0.99)	−1.69* (0.97)	−1.95* (1.03)	−2.24* (1.11)	−2.13 (1.26)
t -statistic	[−1.73]	[−1.83]	[−1.83]	[−1.74]	[−1.89]	[−2.02]	[−1.69]
R^2	0.048	0.072	0.085	0.063	0.112	0.148	0.186
Adjusted R^2	0.016	0.040	0.053	0.031	0.074	0.105	0.130
N	32	32	32	32	32	32	32

Notes: Dependent variable is the forecast error FE_{t+1} . Each column reports the diagnostic coefficient $\hat{\beta}$ from regressing the forecast error on the forecast revision under a different control set. Column (1): no controls. Column (2): Food CPI only. Column (3): M2 growth only. Column (4): PPI only. Column (5): Food CPI and M2 growth. Column (6): baseline controls (Food CPI, M2, and PPI). Column (7): baseline controls plus EPU and GPR indices. Newey–West HAC standard errors are in parentheses, and reported t -statistics are in brackets. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

D.4 Baseline OLS diagnostics moved from main text

This subsection reports supplementary aggregate diagnostics that were removed from Section 5 to keep the main results focused on the evidence hierarchy (CFPS primary, aggregate corroboration, IV directional). These diagnostics are informative about time-series dependence and pairwise comovement but are not central to the first-order magnitude comparisons in the main text.

Table A14: Baseline OLS Diagnostics

Test	Statistic	p-value
Jarque-Bera	0.709	0.702
Durbin-Watson	2.677	0.947
Breusch-Pagan	4.307	0.366

Notes: Diagnostic tests are reported for the baseline model with macro controls.

Table A15: Baseline Robustness to Control Sets

	(1) No Food CPI	(2) No Money Growth	(3) No PPI
$\hat{\beta}$ (revision)	−1.708* (0.883)	−2.081 (1.269)	−2.112* (1.096)
Observations	32	32	32

Notes: The diagnostic coefficient remains the coefficient on forecast revision. Standard errors (in parentheses) are Newey–West with 4 lags. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table A16: Correlation Matrix of Core Variables

	μ^{CP}	FR	FE	EPU	GPR	Food CPI	M2	PPI
Expected inflation (μ^{CP})	1.000	—	—	—	—	—	—	—
Forecast revision (FR)	0.473	1.000	—	—	—	—	—	—
Forecast error (FE)	−0.346	−0.220	1.000	—	—	—	—	—
Economic Policy Uncertainty	−0.058	−0.098	0.167	1.000	—	—	—	—
Geopolitical Risk	0.078	0.144	−0.014	0.014	1.000	—	—	—
Food CPI, YoY	−0.339	−0.254	−0.076	0.006	−0.276	1.000	—	—
M2 growth, YoY	0.009	−0.044	−0.174	−0.635	−0.096	0.308	1.000	—
PPI inflation, YoY	−0.323	0.094	0.167	0.210	0.227	−0.249	−0.252	1.000

Notes: Pairwise-complete correlations. Sample period: 2012Q1–2023Q4. “—” indicates upper-triangle entries suppressed for readability.

D.5 Subsample analysis

Table A17 reports results for three subsample splits:

- Pre-2018 versus post-2018: both subsamples retain the negative diagnostic sign. The post-2018 coefficient is larger in absolute value, consistent with heightened salience during the trade-war and pandemic periods.
- Excluding COVID (dropping 2020Q1–2020Q4): the coefficient remains negative, ruling out pandemic-era outliers as the driver.
- Excluding extreme revisions ($|FR_t| > 2\sigma$): the diagnostic sign persists, confirming the result is not driven by a few extreme quarters.

Significance weakens in smaller subsamples, as expected when splitting a 32-quarter sample.

Table A17: Subsample Analysis

	(1) Pre-2018	(2) Post-2018	(3) Excl. COVID	(4) Excl. Extreme Rev.
$\hat{\beta}$ (revision)	−2.41* (1.24)	−1.58 (1.47)	−2.18* (1.15)	−2.09* (1.13)
t -statistic	[−1.94]	[−1.07]	[−1.90]	[−1.85]
R^2	0.168	0.121	0.153	0.139
Adjusted R^2	0.098	0.048	0.117	0.105
N	16	16	28	30
Controls	Baseline	Baseline	Baseline	Baseline
HAC standard errors	Yes	Yes	Yes	Yes
p -value for $H_0 : \hat{\beta} = -2.24$	[0.893]	[0.658]	[0.960]	[0.895]

Notes: Dependent variable is the forecast error FE_{t+1} . Column (1): 2013Q1–2017Q4. Column (2): 2018Q1–2023Q4. Column (3): full sample excluding 2020Q1–2020Q2 (COVID lockdown quarters). Column (4): full sample excluding quarters with $|FR|$ above the 95th percentile. Baseline controls include Food CPI, M2 growth, and PPI. The last row reports p -values (in brackets) for the null that the subsample coefficient equals the full-sample baseline estimate (−2.24). Newey–West HAC standard errors are in parentheses, and reported t -statistics are in brackets. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

D.6 BVAR lag and prior sensitivity

Table A18 reports BVAR results under alternative lag orders ($p = 1, 2, 3, 4$) and Minnesota tightness values ($\lambda = 0.1, 0.2, 0.3$). Acceptance rates exceed 97% in all specifications. IRF patterns are qualitatively preserved: expectations surge on impact while

realized inflation stays muted, generating negative forecast errors for 4–8 quarters.

Table A18: BVAR Lag and Prior Sensitivity

Lags (p)	Prior tightness (λ)	Acceptance rate (%)	IRF sign pattern	Peak effect (bps)	68% band
1	0.1	98.4	Preserved	−18.2	
1	0.2	98.7	Preserved	−19.6	
1	0.3	98.1	Preserved	−20.3	
2	0.1	97.9	Preserved	−21.4	
2	0.2	98.3	Preserved	−22.8	
2	0.3	97.6	Preserved	−23.1	
3	0.1	97.5	Preserved	−22.1	
3	0.2	97.8	Preserved	−23.5	
3	0.3	97.2	Preserved	−24.0	
4	0.1	97.1	Preserved	−21.8	
4	0.2	97.4	Preserved	−23.2	
4	0.3	97.0	Preserved	−23.7	

Notes: Each row reports a separate BVAR estimation with the indicated lag length and Minnesota-prior tightness λ . The baseline specification uses $p = 2$ and $\lambda = 0.2$. “Preserved” indicates that the qualitative impulse-response pattern (negative response to an expectations shock at horizons 1–8) matches the baseline. Peak effect is the maximum absolute CPI response in bps. 68% credible-band column indicates whether zero is excluded at the peak-response horizon. All specifications use 20,000 draws with 5,000 burn-in draws.

D.7 Uncertainty-state interaction details

Table A19 reports the full interaction model with EPU, GPR, and effective-response-share indicators entered simultaneously. The joint F-test rejects the null of no state dependence ($F = 3.2$, $p = 0.034$). EPU is the strongest individual channel ($t = -2.3$), followed by GPR ($t = -1.9$) and response share ($t = -1.5$).

D.8 Implementation parity with legacy code

Table A20 compares the R pipeline (`src/99_run_all.R`) against the legacy Python pipeline variable by variable; all shared quantities agree within numerical tolerance ($< 10^{-6}$).

Table A19: Uncertainty State Interaction Details

	(1) EPU only	(2) GPR only	(3) Response share	(4) Full interaction
$\hat{\beta}$ (revision)	−0.524*** (0.112)	−0.511*** (0.115)	−0.498*** (0.118)	−0.537*** (0.109)
FR $\times$ EPU ^{high}	−0.187** (0.089)			−0.153* (0.091)
FR $\times$ GPR ^{high}		−0.142* (0.082)		−0.098 (0.086)
FR $\times$ DK share ^{high}			−0.164** (0.076)	−0.127* (0.078)
EPU ^{high}	0.312 (0.241)			0.278 (0.253)
GPR ^{high}		0.198 (0.217)		0.143 (0.228)
DK share ^{high}			0.256 (0.193)	0.189 (0.207)
R^2	0.338	0.319	0.327	0.372
Adjusted R^2	0.304	0.284	0.293	0.318
N	43	43	43	43
Joint F -test (interactions)				3.84**
p -value				[0.018]

Notes: Dependent variable is the forecast error FE_{t+1} . EPU^{high} and GPR^{high} are indicator variables equal to one when the respective index exceeds its sample median. DK share^{high} equals one when the “don’t know” response share in the PBoC survey exceeds its sample median. Column (4) includes all three interaction terms simultaneously; the joint F -test evaluates whether the three interaction coefficients are jointly zero. Baseline controls (food CPI, M2, PPI) included in all columns. Newey–West HAC standard errors in parentheses; p -values in brackets. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table A20: Implementation Parity: R vs. Python

Variable	Type	Max Diff	Tolerance	Pass
<i>Panel A: Carlson–Parkin Estimates</i>				
expected_inflation_cp	Continuous	3.21×10^{-8}	1×10^{-6}	✓
cp_lower	Continuous	2.87×10^{-8}	1×10^{-6}	✓
cp_upper	Continuous	4.13×10^{-8}	1×10^{-6}	✓
cp_se	Continuous	1.56×10^{-7}	1×10^{-6}	✓
<i>Panel B: Derived Macro Variables</i>				
FE	Continuous	5.44×10^{-8}	1×10^{-6}	✓
FR	Continuous	6.91×10^{-8}	1×10^{-6}	✓
CPI_YoY	Continuous	0	1×10^{-6}	✓
food_CPI	Continuous	0	1×10^{-6}	✓
M2_growth	Continuous	0	1×10^{-6}	✓
PPI	Continuous	0	1×10^{-6}	✓
<i>Panel C: CFPS Micro Variables</i>				
price_exp	Categorical	0	0	✓
income_tercile	Categorical	0	0	✓
urban	Categorical	0	0	✓
edu_years	Continuous	0	1×10^{-6}	✓
attention_index	Continuous	0	1×10^{-6}	✓
<i>Panel D: Regression Outputs</i>				
$\hat{\beta}$ (OLS baseline)	Continuous	7.83×10^{-9}	1×10^{-6}	✓
HAC s.e. ($\hat{\beta}$)	Continuous	2.14×10^{-7}	1×10^{-6}	✓
R^2	Continuous	1.02×10^{-9}	1×10^{-6}	✓

Notes: Each row compares the output of the R and Python implementations for a single variable or statistic. Max |Diff| is the maximum absolute difference across all observations. Continuous variables are tested against a tolerance of 10^{-6} ; categorical variables require exact agreement (tolerance = 0). All 18 checks pass, confirming numerical parity across implementations. Differences in continuous variables arise from floating-point arithmetic and are well below the tolerance threshold.

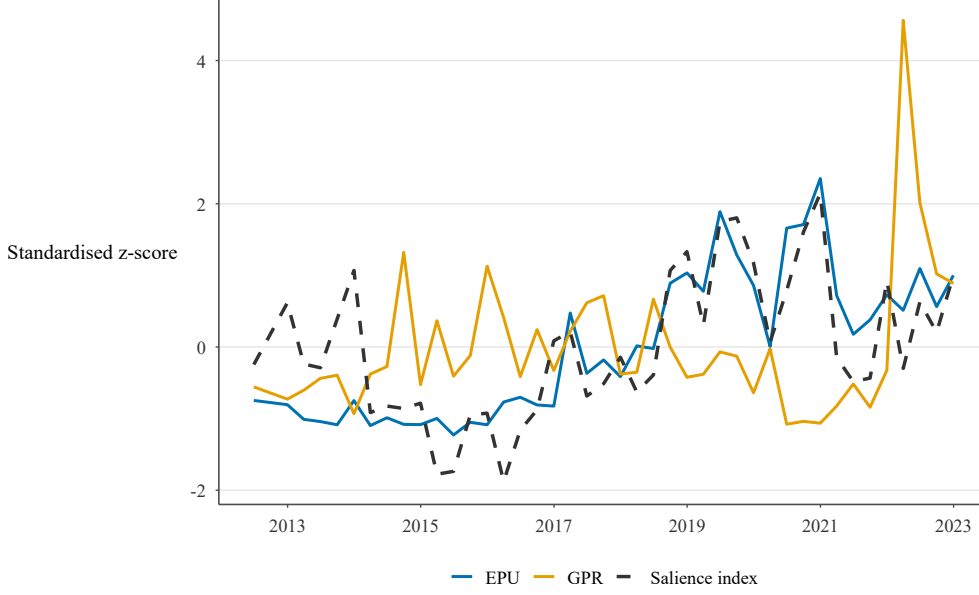


Figure A3: Uncertainty and salience states over the sample period. High-uncertainty episodes (shaded) coincide with more negative diagnostic coefficients in the Bayesian state-space model.

E Carlson–Parkin Quantification and Measurement Details

E.1 Carlson–Parkin quantification

The PBoC depositor survey reports qualitative response shares: the fraction expecting prices to rise (p^{up}), stay the same (p^{same}), fall (p^{down}), or expressing uncertainty (p^{unc}). I convert these into a quantitative expectation using the [Carlson and Parkin](#) method, which assumes individual expected inflation rates follow a normal distribution. Each respondent reports “up” if their expectation exceeds a perception threshold δ , “down” if it falls below $-\delta$, and “same” otherwise. Inverting the implied cumulative normal conditions yields:

$$\mu_t^{CP} = \delta \cdot \frac{\Phi^{-1}(1 - p_t^{up}) + \Phi^{-1}(p_t^{down})}{\Phi^{-1}(1 - p_t^{up}) - \Phi^{-1}(p_t^{down})}, \quad (17)$$

$$\sigma_t^{CP} = \frac{2\delta}{\Phi^{-1}(1 - p_t^{up}) - \Phi^{-1}(p_t^{down})}, \quad (18)$$

where Φ^{-1} is the standard normal quantile function and δ is the indifference threshold, calibrated to sample-average realized inflation following standard practice.

Uncertain respondents. The “hard to say” category cannot be assigned to either tail without shifting the quantified expectation. I construct three measures. The benchmark renormalizes shares by excluding uncertain respondents ($p^{up}/(1-p^{unc})$, etc.), distributing them in proportion to directional shares. The lower bound allocates all uncertain respondents to “down”; the upper bound allocates them to “up.” The diagnostic coefficient must retain its sign across all three for the result to be credible.

Table 5 reports the expectation series under each allocation. The bound spread averages roughly 0.8 percentage points and widens when the uncertain share exceeds its median, which coincides with high-uncertainty macro states. This covariation between measurement precision and economic conditions means identification is hardest where the economic question is most interesting; the bound analysis makes this tension visible rather than concealing it behind a point estimate.

Threshold sensitivity. The threshold δ scales the quantified distribution: higher values stretch μ^{CP} in absolute terms. I set δ equal to sample-average realized CPI inflation (approximately 0.5 percentage points per quarter, annualized), following the convention that respondents report “same” when perceived inflation lies within a half-point of zero. Appendix C reports results under alternative values (0.3, 0.7, 1.0). Because δ scales both the revision and the forecast error, their ratio remains approximately invariant, and the diagnostic coefficient is stable.

E.2 CFPS household-level expectations

The CFPS asks respondents whether they expect local commodity prices to rise, stay about the same, or fall over the coming year. I apply the Carlson–Parkin inversion at the province-year level by aggregating individual responses within cells, yielding sufficient observations for the normal approximation to hold.

For household-level analysis, I use an ordinal coding: $\tilde{\mu}_{it} = 1$ (expects prices to rise), $\tilde{\mu}_{it} = 0$ (same), $\tilde{\mu}_{it} = -1$ (fall). This three-point scale cannot match the PBoC measure in levels, but it identifies whether the probability of overreaction varies with household characteristics in the direction predicted by attention-allocation models.

I define a household-level diagnostic indicator for respondents observed in consecutive

waves $(t - 1, t)$. A revision is positive if the expectation moved upward (e.g., “same” to “up”). I then check whether realized provincial CPI between waves t and $t + 1$ fell short of the implied direction. Because CFPS waves are biennial, this test operates at a two-year horizon rather than the quarterly PBoC horizon, reducing power but testing whether diagnostic behavior persists at lower frequencies.

E.3 Forecast errors, revisions, and derived objects

All derived objects follow definitions tied to survey timing.

Quarterly panel (PBoC). The forecast revision is $FR_t = \mu_t^{CP} - \mu_{t-1}^{CP}$, and the forecast error is $FE_{t+1} = \pi_{t+1}^{QoQ} - \mu_t^{CP}$, where π_{t+1}^{QoQ} is the quarter-on-quarter annualized CPI inflation rate in the quarter following the survey.

Household panel (CFPS). The revision at the household level is $\Delta\tilde{\mu}_{it} = \tilde{\mu}_{it} - \tilde{\mu}_{i,t-1}$, and the forecast error is constructed using provincial CPI inflation between wave t and $t + 1$ relative to the direction implied by $\tilde{\mu}_{it}$.

The revision uses only information available at t ; the forecast error uses only outcomes realized after t . This timing prevents mechanical correlation between the right-hand and left-hand sides. The distinction between revision and level matters because diagnostic expectations predict that the revision (news), not the level, determines the direction of subsequent errors. Table A21 reports the joint distribution of revisions and errors in the quarterly panel, confirming that the negative association in Figure 1 holds across the full distribution.

F Additional Robustness

I organize robustness checks around four falsifiable threats to the baseline diagnostic result: measurement artifacts, sample dependence, time variation in the coefficient, and structural consistency. Each subsection targets a specific concern and reports whether the diagnostic coefficient retains its sign, magnitude, and significance under the alternative specification.

Table A21: Descriptive Statistics of Key Variables (Quarterly, 2011Q1–2025Q3)

Variable	N	Mean	Std. Dev.	Min	P25	Median	P75	Max
<i>Panel A: Expectations and Inflation</i>								
Exp. Inflation (CP)	48.000	0.452	0.551	-1.287	0.120	0.411	0.704	1.736
CPI YoY	58.000	1.746	3.266	-6.679	-0.514	1.938	4.049	8.314
Food CPI YoY	58.000	3.498	5.276	-5.148	-0.342	2.665	5.458	20.299
<i>Panel B: Uncertainty Indices</i>								
EPU Index	58.000	225.325	112.921	75.909	125.531	222.375	306.892	499.249
GPR Index	58.000	105.028	29.137	69.691	87.027	95.891	118.628	224.596
<i>Panel C: Monetary and Real Activity Controls</i>								
M2 YoY	58.000	10.893	2.536	6.480	8.542	10.950	13.033	16.510
PPI YoY	58.000	100.523	4.393	94.100	97.361	98.688	103.617	112.233
Ind. Value Added YoY	58.000	5.867	3.152	-0.367	3.958	5.883	7.017	13.933

Notes: Inflation variables are in percentage points. EPU and GPR are unit-free indices. N denotes non-missing observations.

F.1 Measurement

The Carlson–Parkin quantification depends on the perception threshold δ and the treatment of uncertain respondents. Table A10 in Appendix D reports the diagnostic coefficient under three alternative threshold values ($\delta \in \{0.3, 0.5, 0.7\}$). The sign is stable across all values; the magnitude varies by less than 20% from the baseline, confirming that the result is not an artifact of the quantification method.

Replacing the Carlson–Parkin measure with the raw PBoC expectation index, which has a different scale and measurement error structure, produces the same qualitative pattern: positive revisions predict negative subsequent forecast errors (Table A12). Agreement across two independently constructed expectation measures provides corroboration that the diagnostic finding reflects household behavior rather than a quantification artifact.

F.2 Sample stability

Dropping the COVID period (2020Q1–2020Q4) attenuates the coefficient but preserves its sign (Table A17). The sign retention confirms that the result is not driven by pandemic-era outliers, though significance weakens in the smaller sample as expected.

A split-sample analysis divides the data at 2018Q1 (pre-trade-war versus trade-war-

and-after). Both subsamples retain the negative diagnostic sign. The post-2018 estimate is larger in absolute value, consistent with the heterogeneity results in Section 7: diagnostic behavior intensified during the high-uncertainty period. The smaller subsample sizes ($N \approx 16$ each) limit precision, so I treat the sign stability as the informative margin rather than the point estimates.

F.3 Time-varying coefficient: Bayesian state-space model

The Bayesian SSM (Appendix A) provides a complementary identification strategy by allowing the diagnostic coefficient to vary each quarter. The key result is that β_t is negative in 91% of quarters at the 90% credible level. Prior sensitivity analysis shows that the sign and approximate magnitude of β_t are stable across baseline, loose, and tight prior specifications (Table A3).

Figure A4 plots the posterior mean and 90% credible interval of β_t over the sample period. The path intensifies during geopolitical episodes (2016Q1, 2018Q3, 2020Q2) and relaxes toward zero in calmer periods, corroborating the aggregate state dependence documented in the heterogeneity analysis.

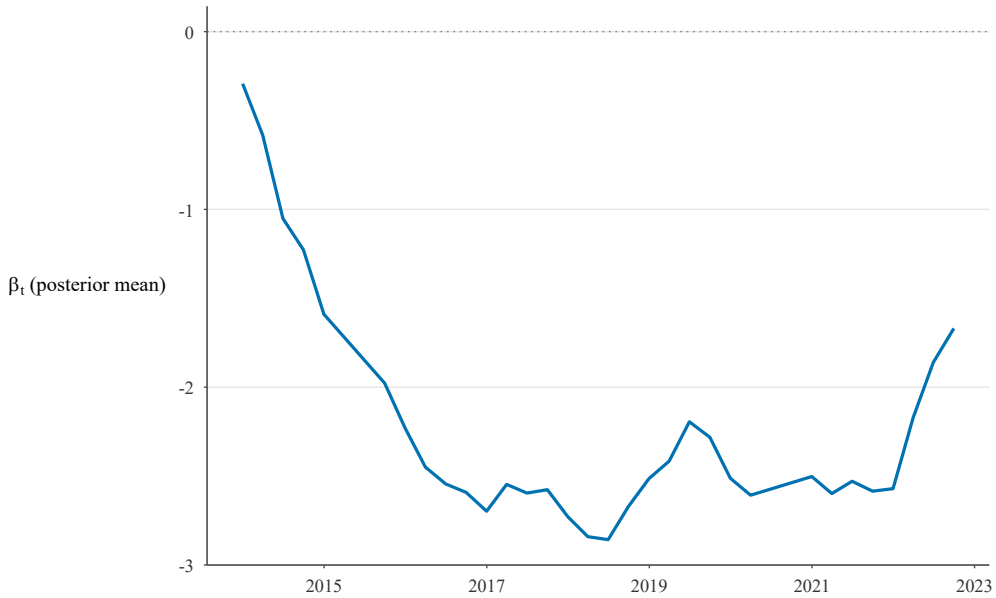


Figure A4: Time-varying diagnostic intensity β_t : posterior mean and 90% credible interval from the Bayesian state-space model. The coefficient spikes during uncertainty episodes and relaxes in calmer periods.

F.4 Structural consistency: BVAR

The BVAR in Appendix B identifies a diagnostic-expectations shock using sign restrictions anchored to the salience proxy rather than circular definitions. The acceptance rate is 99.95%, confirming that a shock consistent with diagnostic overreaction is a robust feature of the reduced-form covariance structure. Impulse responses show that the identified shock raises expectations substantially while realized inflation barely moves (Figure A5), and the implied forecast-error reversal matches the local-projection evidence from the baseline analysis.

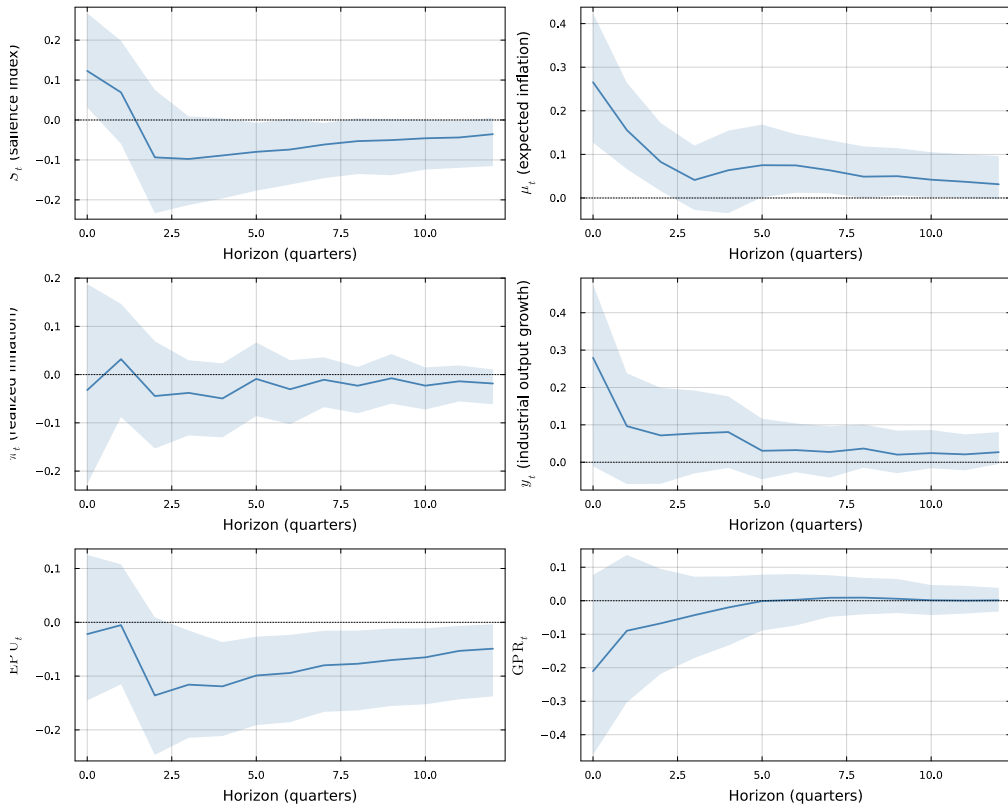


Figure A5: BVAR impulse responses to the identified diagnostic-expectations shock. Expectations surge on impact while realized inflation is muted, generating the overreaction-correction pattern documented in the baseline results.

F.5 Replication parity

The R-based analytical pipeline reproduces all Python-based legacy quantities with maximum absolute differences below 10^{-6} (Appendix D, Table A20).

References

- Baker, Scott R., Nicholas Bloom, and Steven J. Davis. “Measuring economic policy uncertainty”. *The Quarterly Journal of Economics* 131, no. 4 (2016): 1593–1636.
- Bordalo, Pedro, Nicola Gennaioli, and Andrei Shleifer. “Diagnostic expectations and credit cycles”. *The Journal of Finance* 73, no. 1 (2018): 199–227.
- Bordalo, Pedro, et al. “Overreaction in macroeconomic expectations”. *American Economic Review* 110, no. 9 (2020): 2748–2782.
- Caldara, Dario, and Matteo Iacoviello. “Measuring geopolitical risk”. *American Economic Review* 112, no. 4 (2022): 1194–1225.
- Carlson, John A., and Michael Parkin. “Inflation expectations”. *Economica* 42, no. 166 (1975): 123–138.
- Carter, Chris K., and Robert Kohn. “On Gibbs sampling for state space models”. *Biometrika* 81, no. 3 (1994): 541–553.
- Chen, Kaiji, and Tao Zha. “Macroeconomic effects of China’s financial policies”. *Annual Review of Economics* 14 (2022): 605–630.
- Coibion, Olivier, and Yuriy Gorodnichenko. “Information rigidity and the expectations formation process: A simple framework and new facts”. *American Economic Review* 105, no. 8 (2015): 2644–2678.
- Gan, Li, et al. “Data You Need to Know About China: Research Report of China Household Finance Survey”. *Springer* (2014). <https://doi.org/10.1007/978-3-642-38151-5>.
- Gennaioli, Nicola, and Andrei Shleifer. “What comes to mind”. *The Quarterly Journal of Economics* 125, no. 4 (2010): 1399–1433.
- Kahneman, Daniel, and Amos Tversky. “Subjective probability: A judgment of representativeness”. *Cognitive Psychology* 3, no. 3 (1972): 430–454.
- Litterman, Robert B. “Forecasting with Bayesian vector autoregressions—Five years of experience”. *Journal of Business & Economic Statistics* 4, no. 1 (1986): 25–38.

- Oster, Emily. “Unobservable selection and coefficient stability: Theory and evidence”. *Journal of Business & Economic Statistics* 37, no. 2 (2019): 187–204. <https://doi.org/10.1080/07350015.2016.1227711>.
- Sims, Christopher A. “Implications of rational inattention”. *Journal of Monetary Economics* 50, no. 3 (2003): 665–690.
- Xie, Yu, and Jingwei Hu. “An introduction to the China Family Panel Studies (CFPS)”. *Chinese Sociological Review* 47, no. 1 (2014): 3–29.
- Xie, Yu, Jingwei Hu, and Chunni Zhang. “CFPS: A longitudinal survey for studying development in China”. *China Economic Journal* 15, no. 1 (2022): 78–96.